

A Project Report on
**“QR Code Based Digital Certificate Verification System to
Prevent Forgery”**

Submitted in Partial Fulfillment of the Requirements For the award of the degree

Bachelor of Computer Applications
OF
PRESIDENCY COLLEGE (AUTONOMOUS)

SUBMITTED BY

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23C01074



PRESIDENCY COLLEGE (AUTONOMOUS)

Kempapura, Hebbal, Bengaluru – 24

Re-accredited by NAAC with ‘A+’ & NIRF Ranked

DEPARTMENT OF COMPUTER APPLICATIONS

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CERTIFICATE

This is to certify that **J.Yogitha** with Register No. **23C01074** has satisfactorily completed the Sixth Semester BCA Project titled “ **QR Code Based Digital Certificate Verification System to Prevent Forgery**“, as a partial fulfilment of the requirements for the award of the Degree in **Bachelor of Computer Applications**, awarded by Presidency College (AUTONOMOUS), during the Academic Year **2025-2026**.

Project Guide : Ms. Anu.D

Head of Department
(Department of computer Application)

Examiners

1. _____

2. _____

Reg No: 23C01074

Examination Center: _____

Date of the exam: _____

Declaration

The project titled “**QR CODE BASED DIGITAL CERTIFICATE VERIFICATION SYSTEM TO PREVENT FORGERY**” developed by me in the partial fulfillment of Presidency College (AUTONOMOUS). It is a systematic work carried by us under the guidance of **Ms. Anu.D**, Assistant professor, Department of Computer Applications.

I, declare that this same project has not been submitted to any degree or diploma to the Presidency College (AUTONOMOUS) or any other Universities.

Name of the student: - **J.Yogitha**

Date:- **07/04/2026**

Signature

Acknowledgement

The development of software is generally bit complex and time-consuming task. The goal of developing the project “**QR CODE BASED DIGITAL CERTIFICATE VERIFICATION SYSTEM TO PREVENT FORGERY**” could not be achieved without the encouragements of kindly helpful and supportive people. Here by we convey our sincere thanks for all of them.

I take this opportunity to express my gratitude to people who had been instrumental in the successful completion of this project.

I am thankful to our management trustee for providing us an opportunity to work and complete the project successfully.

I wish to express my thanks to our Principal **Dr. Pradeep Kumar Shinde** for his support to the project work. I would like to acknowledge my gratitude to our HOD of Department of Computer Applications **Dr. Alli A** for her encouragement and support. Without her encouragement and guidance this project would not have materialized.

The guidance and support received from our Internal Guide Ms. Anu.D who contributed to this project, was vital for the success of the project. We are grateful for their constant support and help.

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ABSTRACT

1. ABSTRACT:

The growing use of digital certificates in educational institutions has increased the risk of certificate forgery and unauthorized modifications. Fake certificates can lead to misrepresentation of academic qualifications and reduce trust in institutional credentials. Therefore, it is important to provide a secure and reliable method to verify the authenticity of certificates.

The project titled **“QR Code Based Digital Certificate Verification System to Prevent Forgery”** is designed to provide a secure platform for generating, managing, and verifying digital certificates. In this system, certificates are issued with a unique identification number and an embedded QR code that allows quick verification of the certificate. When the QR code is scanned, it redirects the user to a verification page where the authenticity of the certificate can be checked.

To enhance security, the system also uses a hashing mechanism to ensure that the certificate file has not been altered after it is issued. During verification, the system compares the stored hash value with the newly generated hash value of the certificate file. If both values match, the certificate is considered valid; otherwise, it indicates that the certificate has been tampered with.

The system also provides administrative control to manage student records, generate certificates, upload existing certificates, and revoke certificates when necessary. Additionally, verification activities are recorded to maintain a log of certificate validation attempts.

By integrating QR code verification and secure validation techniques, the proposed system helps institutions prevent certificate forgery, ensures transparency, and simplifies the process of certificate authentication for organizations and individuals.

INTRODUCTION

2. INTRODUCTION:

In recent years, digital certificates have become widely used in educational institutions to recognize academic achievements, course completions, and participation in various activities. However, with the increasing use of digital certificates, the risk of certificate forgery and unauthorized modification has also increased. Fake certificates can be easily created or altered, which may lead to misuse of academic credentials and loss of trust in institutional records. Therefore, it is important to develop a secure system that ensures the authenticity and integrity of issued certificates.

The **QR Code Based Digital Certificate Verification System to Prevent Forgery** is designed to provide a reliable and efficient solution for managing and verifying digital certificates. The system allows institutions to generate certificates with a unique identification number and an embedded QR code. When the QR code is scanned, it directs the user to a verification page where the certificate details can be validated.

The system also incorporates a hashing mechanism to detect any tampering with the certificate file. By comparing stored certificate data with the verification data, the system can determine whether a certificate is valid, revoked, or modified. This helps institutions maintain transparency and prevents the misuse of forged certificates. The system also provides administrative control for managing student records, certificate generation, and certificate verification logs.

Overall, the proposed system simplifies the certificate verification process and ensures that digital certificates remain secure, trustworthy, and easy to validate.

2.1 OBJECTIVE:

- To develop a secure system for generating and managing digital certificates.
- To provide quick and reliable certificate verification using QR codes.
- To prevent certificate forgery by detecting unauthorized modifications.
- To allow administrators to manage certificate records and revoke certificates when necessary.
- To enable students to access and verify their certificates through a dedicated portal.
- To maintain verification logs for tracking certificate validation activities.

2.2 MAJOR MODULES:

1. Admin Module
2. Student Management Module
3. Certificate Generation Module
4. QR Code Generation Module
5. Certificate Verification Module
6. Tamper Detection Module
7. Certificate Revocation Module
8. Verification Log Module

1. ADMIN MODULE:

- Provides full control over the system.
- Allows the administrator to manage student records.
- Enables certificate generation and external certificate uploads.
- Embeds QR codes into certificates.
- Allows the administrator to revoke certificates.
- Displays verification logs and system activities.

2. STUDENT MANAGEMENT MODULE:

- Allows the administrator to add student details.

- Stores student information such as name, register number, course, academic year, and institution.
- Maintains student records in the database.
- Provides data required for certificate generation and verification.

3. CERTIFICATE GENERATION MODULE:

- Allows the administrator to generate digital certificates.
- Collects required student information for certificate creation.
- Generates a certificate with a unique certificate ID.
- Stores certificate details securely in the database.

4. QR CODE GENERATION MODULE:

- Automatically generates a QR code for each certificate.
- Embeds the QR code into the certificate document.
- Stores the QR code image in the system.
- Links the QR code to the certificate verification page.

5. CERTIFICATE VERIFICATION MODULE:

- Allows users to verify certificates by scanning the QR code.
- Retrieves certificate details from the database.
- Displays certificate information on the verification page.
- Shows the verification result as valid, revoked, or tampered.

6. TAMPER DETECTION MODULE:

- Generates a hash value for each certificate file.
- Stores the generated hash value in the database.
- Recalculates the hash value during verification.
- Compares stored and generated hash values to detect modifications.

7. CERTIFICATE REVOCATION MODULE:

- Allows the administrator to revoke certificates when required.
- Updates the certificate status in the system.
- Prevents revoked certificates from being considered valid.
- Displays revoked status during verification.

8. VERIFICATION LOG MODULE:

- Records each certificate verification attempt.
- Stores certificate ID and scan date.
- Saves the verification result in the database.
- Maintains a history of verification activities for monitoring.

SYSTEM ANALYSIS

3. SYSTEM ANALYSIS:

System analysis is the process of studying and understanding the current system, identifying its limitations, and proposing an improved system to overcome the existing problems. In the context of certificate management, traditional methods of issuing and verifying certificates have several challenges such as forgery, lack of quick verification, and difficulty in maintaining records. Therefore, a more secure and efficient system is required to ensure the authenticity and integrity of certificates.

3.1 EXISTING SYSTEM:

In the existing system, certificates are generally issued in printed or simple digital formats without strong verification mechanisms. Verification of certificates often requires manual checking by institutions or organizations, which is time-consuming and inefficient.

➤ LIMITATIONS OF THE EXISTING SYSTEM:

- Certificates can be easily forged or duplicated.
- Verification of certificates requires manual checking.
- There is no quick method to verify certificate authenticity.
- Maintaining records of issued certificates is difficult.
- Institutions may face difficulty tracking certificate misuse.
- There is no automated system to detect certificate tampering.

3.2 PROPOSED SYSTEM:

The proposed system introduces a **QR Code Based Digital Certificate Verification System** to provide a secure and efficient solution for managing and verifying certificates. In this system, each certificate is generated with a unique certificate ID and an embedded QR code that allows instant verification.

When the QR code is scanned, it redirects the user to a verification page where certificate details are validated. The system also uses a hashing mechanism to ensure that the certificate file has not been modified after issuance. If any modification is detected, the certificate is

marked as tampered. Additionally, the system allows administrators to revoke certificates and maintain verification logs.

➤ **ADVANTAGES OF THE PROPOSED SYSTEM:**

- Provides secure digital certificate management.
- Enables quick verification through QR code scanning.
- Detects certificate tampering using hash validation.
- Reduces the risk of certificate forgery.
- Allows administrators to manage and revoke certificates.
- Maintains verification logs for monitoring and auditing.

The proposed system improves transparency, reliability, and efficiency in certificate verification while ensuring that only authentic certificates are accepted.

REQUIREMENT ANALYSIS

4. REQUIREMENT ANALYSIS:

Requirement analysis identifies the system requirements needed for the successful development and functioning of the proposed system. It describes what the system should do and the conditions under which it should operate. The requirements are mainly categorized into **functional requirements** and **non-functional requirements**.

4.1 FUNCTIONAL REQUIREMENTS:

Functional requirements define system operations

- The system must allow the administrator to log in securely.
- The system must allow the administrator to add and manage student details.
- The system must allow the administrator to generate digital certificates for students.
- The system must generate a unique certificate ID for each certificate.
- The system must generate and embed a QR code in each certificate.
- The system must store certificate details and student information in the database.
- The system must allow administrators to upload external certificates.
- The system must allow administrators to revoke certificates when necessary.
- The system must allow students to view and download their certificates.
- The system must allow users to verify certificates by scanning the QR code.
- The system must detect certificate tampering using a hash verification method.
- The system must record verification details such as certificate ID, scan date, and verification result.

4.2 NON-FUNCTIONAL REQUIREMENTS:

Non-functional requirements define system quality and performance.

- The system should provide secure access to administrative functions.
- The system should ensure data integrity and prevent unauthorized modifications.
- The system should provide fast and efficient certificate verification.
- The system should maintain accurate and reliable database records.
- The system should be reliable and available whenever verification is required.
- The system should maintain logs for monitoring and auditing purposes.

4.3 HARDWARE AND SOFTWARE REQUIREMENTS:

HARDWARE	
Processor	Intel i3 or above
RAM	Minimum 4 GB
Storage	500 MB free space
Device	Laptop/Desktop
Mobile device	Any smartphone with QR scanner
SOFTWARE	
Operating System	Windows 10/11
Server	XAMPP (Apache + MySQL)
Frontend	HTML, CSS
Backend	PHP (phpMyAdmin)
Database	MySQL
PDF Library	FPDF, FPDF
QR Code Generation	QR API / PHP QR Code
IDE/Editor	Visual Studio Code
Browser	Google Chrome or Edge

Table 4.3.1

SYSTEM DESIGN AND SPECIFICATIONS

5. SYSTEM DESIGN AND SPECIFICATIONS:

5.1 ER DIAGRAM:

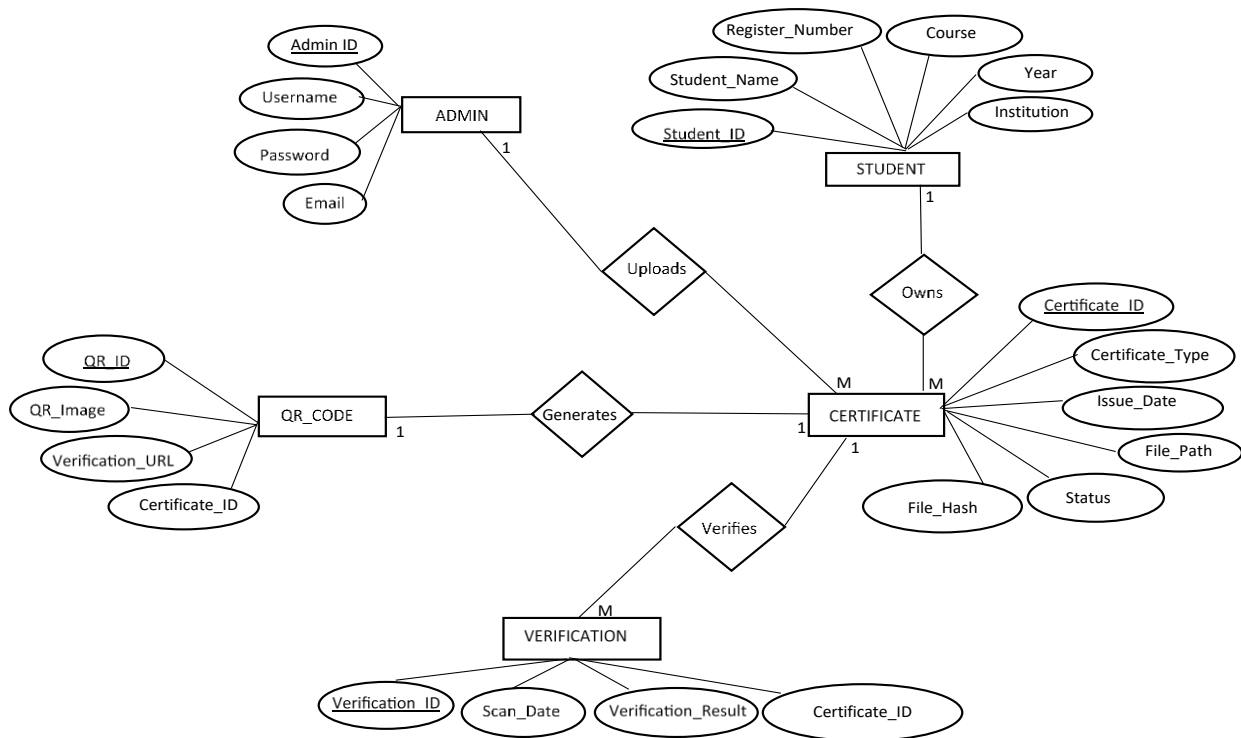


Fig 5.1.1

5.2 DATA FLOW DIAGRAM:

LEVEL 0:

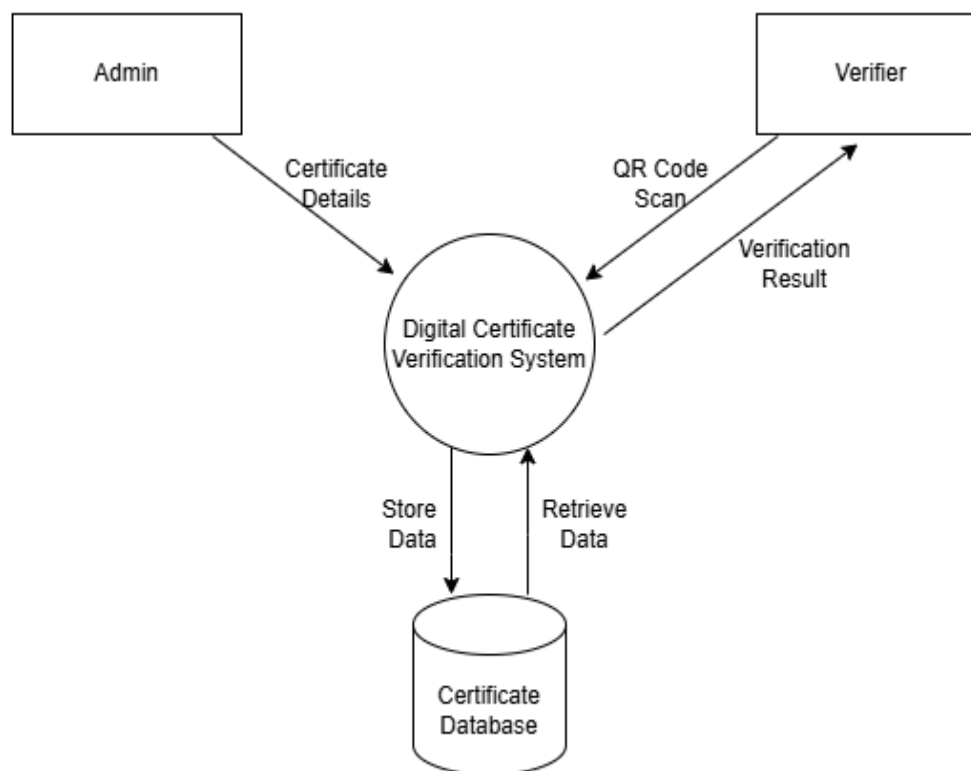


Fig 5.2.1

LEVEL 1:

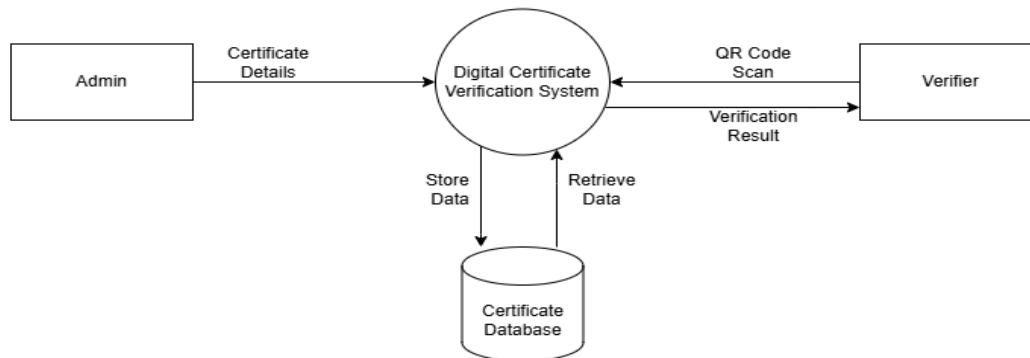


Fig 5.2.2

LEVEL 2:

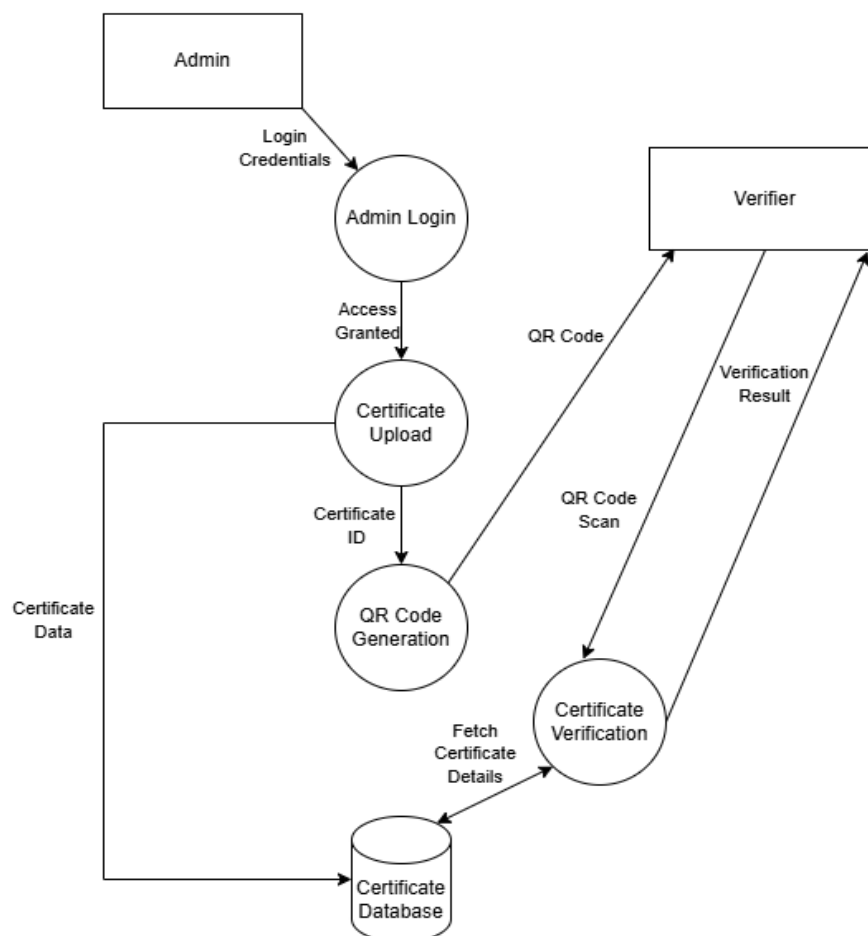


Fig 5.2.3

LEVEL 3:

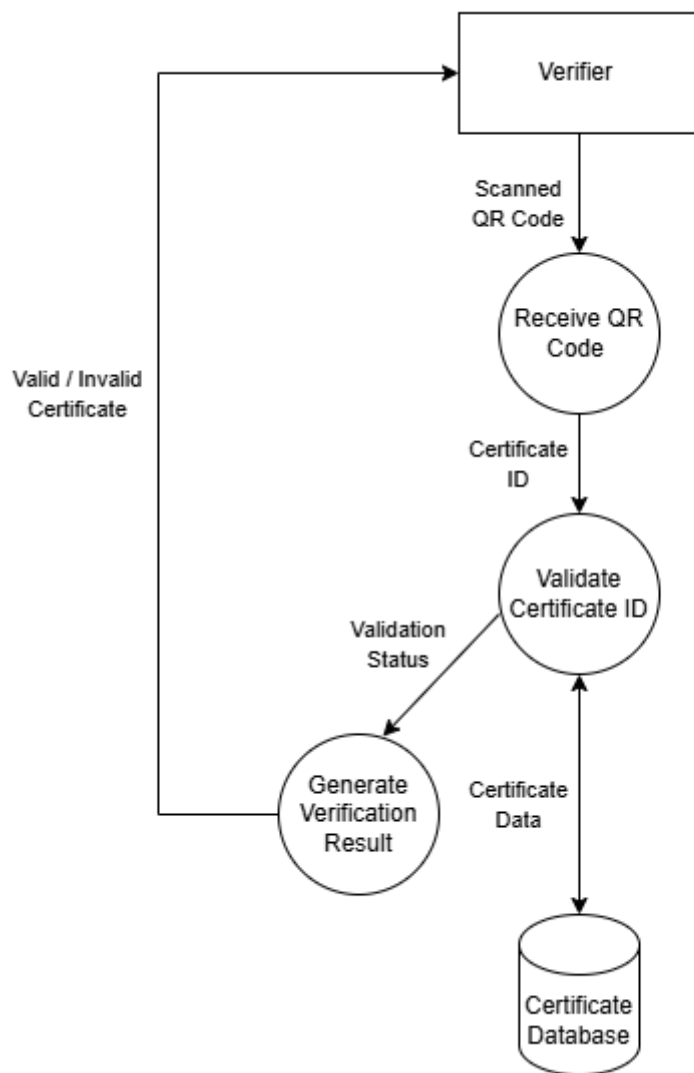


Fig 5.2.4

DATABASE DESIGN

6. DATABASE DESIGN:

6.1 SCHEMA:

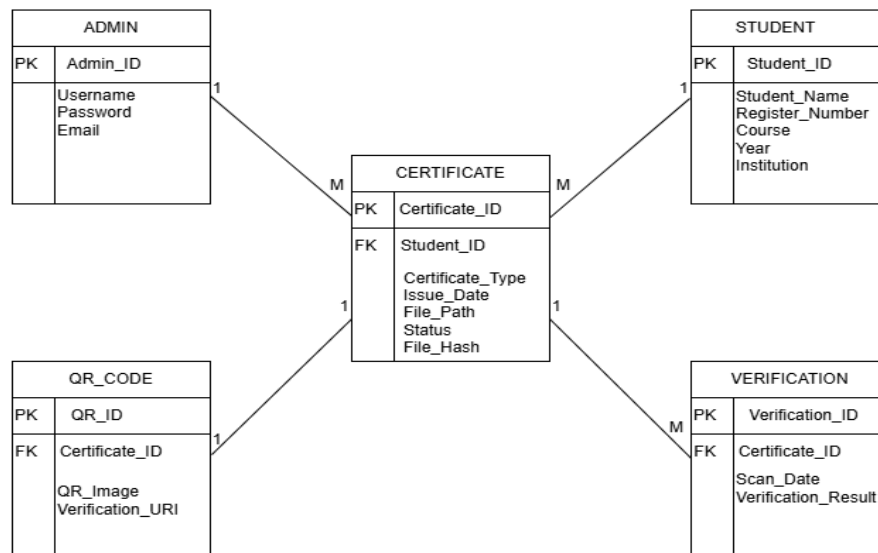


Fig 6.1.1

TESTING AND IMPLEMENTATION

7. TESTING AND IMPLEMENTATION:

1. UNIT TESTING:

Unit testing is used to test individual components of the system independently to ensure that each module functions correctly.

In this project, unit testing was performed on the following modules:

- **Database Connection Module (db.php):** Verified whether the system successfully connects to the MySQL database.
- **Admin Login Module:** Checked whether valid admin credentials allow login and invalid credentials are rejected.
- **Certificate Generation Module:** Tested whether certificates are generated correctly with student details and stored in the database.
- **QR Code Generation Module:** Verified that the system generates a QR code containing the verification URL.
- **SHA-256 Hash Generation:** Tested whether the certificate file hash is correctly generated and stored in the database.

Unit testing ensured that each individual component of the system operates as expected before integrating them together.

2. INTEGRATION TESTING:

Integration testing verifies whether different modules of the system work together correctly when combined.

In this project, the following integrations were tested:

- **Admin Module with Database:** Verified that admin actions such as certificate generation and revocation correctly update database records.
- **Certificate Generation with QR Code Module:** Ensured that the generated certificate contains the embedded QR code linked to the verification page.
- **Verification Module with Hash Validation:** Checked whether the system correctly compares the stored SHA-256 hash with the recalculated file hash.

- **Verification Module with Log System:** Confirmed that every certificate verification attempt is recorded in the verification logs.

Integration testing confirmed that all modules interact properly and the overall system functions smoothly.

3. USER ACCEPTANCE TESTING(UAT):

User Acceptance Testing ensures that the system meets the requirements of the intended users and performs according to their expectations.

In this project, UAT was conducted by testing the system from different user perspectives:

- **Admin User:** Tested certificate generation, certificate revocation, and viewing verification logs.
- **Student User:** Tested student login and certificate viewing functionality.
- **Public User:** Tested QR code scanning to verify certificate authenticity without login.

The results confirmed that the system operates correctly and fulfills the intended requirements for certificate management and verification.

4. PERFORMANCE TESTING:

Performance testing evaluates how efficiently the system performs under normal operating conditions.

In this project, performance testing focused on:

- Speed of certificate generation and PDF creation.
- Time required to generate QR codes.
- Response time of the verification page when scanning QR codes.
- Database retrieval speed for certificate verification.

The system demonstrated efficient performance with quick certificate generation and fast verification results.

5. SECURITY TESTING:

Security testing ensures protection of data and prevents unauthorized access or tampering.

In this project, the following security measures were tested:

- **Admin authentication** ensures that only authorized administrators can generate or revoke certificates.
- **SHA-256 hashing** detects any modification to the certificate file.
- **Certificate revocation mechanism** allows administrators to invalidate certificates when required.
- **Data masking during public verification** protects sensitive student information such as full name and register number.

Security testing confirmed effective protection of certificate data and prevention of forgery.

7.1 TEST CASES:

1. ADMIN LOGIN TEST CASES:

Test Case ID	Test Description	Input	Expected Output	Result
AL01	Login with valid credentials	Correct username & password	Login successful	Pass
AL02	Login with wrong password	Correct username, wrong password	Login failed message	Pass
AL03	Login with wrong username	Wrong username	Login failed message	Pass
AL04	Login with empty username	Empty username	Error displayed	Pass
AL05	Login with empty password	Empty password	Error displayed	Pass
AL06	Login with both empty	No input	Login blocked	Pass
AL07	Login with special characters	@#\$ input	Login denied	Pass
AL08	SQL injection attempt	' OR '1'='1	Login denied	Pass

Table 7.1.1

2. CERTIFICATE GENERATION TEST CASES:

Test Case ID	Test Description	Input	Expected Output	Result
CG01	Generate certificate valid data	All fields filled	Certificate generated	Pass
CG02	Existing student certificate	Existing reg no	Certificate generated	Pass
CG03	Duplicate certificate	Same type	Error message	Pass
CG04	Missing student name	Empty name	Error shown	Pass
CG05	Missing register number	Empty reg no	Error shown	Pass
CG06	Missing course	Empty field	Error shown	Pass
CG07	Check certificate ID format	Generate cert	Unique ID created	Pass
CG08	Check PDF creation	Generate cert	PDF saved	Pass
CG09	Check DB entry	Generate cert	Data stored	Pass
CG10	Large input handling	Long name	Works properly	Pass

Table 7.1.2

3. QR CODE GENERATION TEST CASES:

Test Case ID	Test Description	Input	Expected Output	Result
QR01	Generate QR code	Generate cert	QR created	Pass
QR02	QR image storage	Generate cert	Stored in folder	Pass
QR03	Scan QR code	Mobile scan	Opens page	Pass
QR04	Correct URL in QR	Scan QR	Correct link	Pass
QR05	QR embedded in PDF	Open PDF	QR visible	Pass
QR06	QR readability	Scan QR	Works properly	Pass
QR07	QR size validation	Generate cert	Proper size	Pass
QR08	QR for multiple certs	Multiple gen	Unique QR	Pass
QR09	QR broken link test	Modify URL	Error page	Pass
QR10	QR API availability	Generate QR	Works correctly	Pass

Table 7.1.3

4. CERTIFICATE REVOCATION TEST CASES:

Test Case ID	Test Description	Input	Expected Output	Result
CR01	Revoke valid certificate	Click revoke	Status updated	Pass
CR02	Verify revoked status	Check cert	Revoked shown	Pass
CR03	Download revoked cert	Try download	Blocked	Pass
CR04	Revoke invalid ID	Wrong ID	Error shown	Pass
CR05	Revoke twice	Same cert	No change	Pass
CR06	Verify revoked via QR	Scan QR	Invalid shown	Pass
CR07	DB update check	Revoke cert	DB updated	Pass
CR08	UI update	Refresh page	Status changed	Pass
CR09	Revoke multiple certs	Multiple actions	All revoked	Pass
CR10	Access revoked cert	Open link	Access denied	Pass

Table 7.1.4

5. EXTERNAL CERTIFICATE UPLOAD TEST CASES:

Test Case ID	Test Description	Input	Expected Output	Result
EX01	Upload valid PDF	PDF file	Success	Pass
EX02	Upload invalid format	JPG/PNG	Error	Pass
EX03	QR embed check	Upload	QR added	Pass
EX04	DB entry check	Upload	Stored	Pass
EX05	Large file upload	Big PDF	Works	Pass
EX06	Empty upload	No file	Error	Pass
EX07	File overwrite check	Same file	Unique save	Pass
EX08	QR link working	Scan QR	Opens page	Pass
EX09	Hash generation	Upload	Hash stored	Pass
EX10	External student creation	Upload	Student created	Pass

Table 7.1.5

6. STUDENT VERIFICATION TEST CASES:

Test Case ID	Test Description	Input	Expected Output	Result
SV01	Correct details	Valid input	Certificate shown	Pass
SV02	Wrong details	Invalid input	Error message	Pass
SV03	Empty fields	No input	Error	Pass
SV04	Case sensitivity	Upper/lower	Works	Pass
SV05	Multiple cert display	Valid user	All certs shown	Pass
SV06	Summary display	Valid user	Summary shown	Pass
SV07	Download valid cert	Click download	Success	Pass
SV08	Download revoked cert	Click	Blocked	Pass
SV09	Tampered cert check	Modified file	Tampered	Pass
SV10	Data display accuracy	Valid input	Correct data	Pass

Table 7.1.6

7. PUBLIC VERIFICATION TEST CASES:

Test Case ID	Test Description	Input	Expected Output	Result
PV01	Scan QR	QR scan	Page opens	Pass
PV02	Valid cert verify	Valid ID	Verified	Pass
PV03	Revoked cert verify	Revoked ID	Invalid	Pass
PV04	Invalid ID	Wrong ID	Error	Pass
PV05	Direct URL access	URL open	Works	Pass
PV06	Multiple scans	Repeated scan	Works	Pass
PV07	Slow network	Scan QR	Loads properly	Pass
PV08	Data masking	Scan QR	Masked data	Pass
PV09	Download option	Valid cert	Available	Pass
PV10	External cert verify	Scan QR	Works	Pass

Table 7.1.7

8. TAMPER DETECTION TEST CASES:

Test Case ID	Test Description	Input	Expected Output	Result
TD01	Original file	No change	Valid	Pass
TD02	Modified file	Edited PDF	Tampered	Pass
TD03	Hash mismatch	Change file	Detected	Pass
TD04	Recheck hash	Verify again	Same result	Pass
TD05	Partial edit	Minor change	Detected	Pass
TD06	Rename file	Name change	Valid	Pass
TD07	Replace content	Change data	Tampered	Pass
TD08	Copy file	Duplicate	Valid	Pass
TD09	Missing file	Delete file	Error	Pass
TD10	Corrupt file	Broken PDF	Error	Pass

Table 7.1.8

9. VERIFICATION LOGS TEST CASES:

Test Case ID	Test Description	Input	Expected Output	Result
VL01	QR scan log	Scan QR	Entry created	Pass
VL02	Date record	Scan QR	Date stored	Pass
VL03	Result record	Scan QR	Result saved	Pass
VL04	Multiple scans	Repeat scan	Multiple entries	Pass
VL05	Invalid scan	Wrong ID	Logged	Pass
VL06	Tampered scan	Modified cert	Logged	Pass
VL07	DB storage	Scan	Stored	Pass
VL08	Order check	View logs	Latest first	Pass
VL09	Data accuracy	Scan	Correct data	Pass
VL10	Log display	Open logs	Visible	Pass

Table 7.1.9

10. SECURITY TEST CASES:

Test Case ID	Test Description	Input	Expected Output	Result
SC01	Mask student data	Scan QR	Masked	Pass
SC02	Unauthorized access	Direct URL	Blocked	Pass
SC03	Tamper detection	Modify file	Detected	Pass
SC04	Secure login	Admin login	Protected	Pass
SC05	SQL injection	Input attack	Prevented	Pass
SC06	Data integrity	Verify file	Maintained	Pass
SC07	Access control	Admin pages	Restricted	Pass
SC08	File protection	Cert file	Safe	Pass
SC09	Hash security	Verify hash	Works	Pass
SC10	Session handling	Login session	Maintained	Pass

Table 7.1.10

7.2 CODING:

DATABASE CONNECTION MODULE:

db.php:

```
<?php
$conn = mysqli_connect("localhost", "root", "", "digital_certificate_db");
if (!$conn) {
    die("Database connection failed");
}
?>
```

db_test.php:

```
<?php
include "db.php";
echo "DB CONNECTED";
?>
```

ADMIN LOGIN MODULE:

admin_login.php:

```
<!DOCTYPE html>
<html>
<head>
<style>
  body {
    background-image: url("1.png");
    background-size: cover;
    background-position: center;
    background-repeat: no-repeat;
    height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
  }
  .container {
    background: rgb(253, 253, 255);
    padding: 30px;
    border-radius: 12px;
    box-shadow: 0 0 12px gray;
    text-align: center;
    width: 380px;
  }
  input {
    width: 90%;
    padding: 8px;
  }
  .login-btn {
    width: 90%;
    padding: 10px;
```

```
background-color: green;
color: white;
border: none;
border-radius: 6px;
font-size: 16px;
cursor: pointer;
}
.login-btn:hover {
background-color: darkgreen;
}
</style>
</head>
<body>
<div class="container">
<h2>Admin Login</h2>
<form action="admin_login_check.php" method="POST">
Username:<br>
<input type="text" name="username" required><br><br>
Password:<br>
<input type="password" name="password" required><br><br>
<button type="submit" class="login-btn">Login</button>
</form>
</div>
</body>
</html>
```

admin_login_check.php:

```
<?php
include "../db.php";
$username = $_POST['username'];
$password = $_POST['password'];
$query = "SELECT * FROM ADMIN
        WHERE Username='$username' AND Password='$password'";
$result = mysqli_query($conn, $query);
?>
<!DOCTYPE html>
<html>
<head>
<style>
    body {
        background-image: url("1.png");
        background-size: cover;
        background-position: center;
        background-repeat: no-repeat;
        height: 100vh;
        display: flex;
        justify-content: center;
        align-items: center;
        font-family: Arial, sans-serif;
    }
    .container {
        background: rgba(255, 255, 255, 0.9);
        padding: 30px;
        border-radius: 12px;
        box-shadow: 0 0 12px gray;
        text-align: center;
        width: 380px;
    }
```

```
.link-btn {
    display: inline-block;
    margin-top: 15px;
    padding: 10px 15px;
    background-color: #2ecc71;
    color: white;
    text-decoration: none;
    border-radius: 6px;
    font-weight: bold;
}

.link-btn:hover {
    background-color: #27ae60;
}

</style>
</head>
<body>
<div class="container">
    <?php
    if (mysqli_num_rows($result) == 1) {
        echo "<h3 style='color:green;'>Login successful</h3>";
        echo "<a class='link-btn' href='admin_home.php'>
        Go to Generate Certificate
        </a>";
    } else {
        echo "<h3 style='color:red;'>Invalid login</h3>";
        echo "<a class='link-btn' href='admin_login.html'>
        Try Again
        </a>";
    }
    ?>
</div>
</body></html>
```

ADMIN DASHBOARD MODULE:

admin_home.php:

```
<!DOCTYPE html>
<html>
<head>
<style>
  body {
    background-image: url("1.png");
    background-size: cover;
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
  }
  .box {
    background: white;
    padding: 40px;
    border-radius: 10px;
    text-align: center;
    box-shadow: 0 0 10px gray;
  }
  .btn {
    display: block;
    width: 250px;
    padding: 12px;
    margin: 15px;
    background: #2ecc71;
    color: white;
    text-decoration: none;
    border-radius: 6px;
    font-size: 16px;
```

```

    }
    .btn2 {
        background: #3498db;
    }
</style>
</head>
<body>
    <div class="box">
        <h2>Admin Panel</h2>
        <a class="btn" href="generate_certificate.php">
            Generate Certificate (College Details)
        </a>
        <a class="btn btn2" href="upload_external.php">
            Upload External Certificate
        </a>
    </div>
</body> </html>

```

admin_dashboard.php:

```

<?php
include "../db.php";
$total_cert = mysqli_fetch_assoc(mysqli_query(
    $conn,
    "SELECT COUNT(*) AS total FROM CERTIFICATE"
));
$valid_cert = mysqli_fetch_assoc(mysqli_query(
    $conn,
    "SELECT COUNT(*) AS total FROM CERTIFICATE WHERE Status='Valid'"
));
$revoked_cert = mysqli_fetch_assoc(mysqli_query(
    $conn,
    "SELECT COUNT(*) AS total FROM CERTIFICATE WHERE Status='Revoked'"
));

```

```
$total_verify = mysqli_fetch_assoc(mysqli_query(
    $conn,
    "SELECT COUNT(*) AS total FROM VERIFICATION"
));
?>
<!DOCTYPE html>
<html>
<head>
    <title>Admin Dashboard</title>
    <style>
        body {
            background-image: url("1.png");
            background-size: cover;
            background-position: center;
            background-repeat: no-repeat;
            font-family: Arial;
            text-align: center;
            padding-top: 40px;
        }
        .box {
            display: inline-block;
            background: white;
            padding: 25px;
            margin: 15px;
            border-radius: 10px;
            box-shadow: 0 0 10px gray;
            width: 200px;
        }
        h2 {
            margin-bottom: 30px;
        }
    </style>
```



```
</head>
<body>
  <h2>Admin Dashboard</h2>
  <div class="box">
    <h3>Total Certificates</h3>
    <p><?php echo $total_cert['total']; ?></p>
  </div>
  <div class="box">
    <h3>Valid Certificates</h3>
    <p><?php echo $valid_cert['total']; ?></p>
  </div>
  <div class="box">
    <h3>Revoked Certificates</h3>
    <p><?php echo $revoked_cert['total']; ?></p>
  </div>
  <div class="box">
    <h3>Total Verifications</h3>
    <p><?php echo $total_verify['total']; ?></p>
  </div>
</body>
</html>
```

CERTIFICATE GENERATION MODULE:

generate_certificate.php:

```
<?php
error_reporting(E_ALL);
ini_set('display_errors', 1);
include "../db.php";
require "../fpdf/fpdf.php";
require "../fpdi/src/autoload.php";
use setasign\Fpdi\Fpdi;

$message = "";
$showPopup = false;
$download_link = "";
if (isset($_POST['revoke'])) {
    $register_number = $_POST['register_number'];
    $student = mysqli_query($conn, "
SELECT Student_ID FROM STUDENT
WHERE Register_Number='$register_number'
");
    if (mysqli_num_rows($student) > 0) {
        $row = mysqli_fetch_assoc($student);
        $student_id = $row['Student_ID'];
        mysqli_query($conn, "
UPDATE CERTIFICATE
SET Status='Revoked'
WHERE Student_ID='$student_id'
AND Certificate_Type='{$_POST['certificate_type']}'
");
        $message = "Certificate Revoked Successfully";
        $showPopup = true;
    } else {
        $message = "Register Number Not Found";
    }
}
```

```

        $showPopup = true;
    }
}
if (isset($_POST['generate'])) {
    $student_name = strtoupper($_POST['student_name']);
    $register_number = $_POST['register_number'];
    $course = strtoupper($_POST['course']);
    $year = $_POST['year'];
    $institution = $_POST['institution'];
    $certificate_type = $_POST['certificate_type'];
    $check_student = mysqli_query(
        $conn,
        "SELECT      Student_ID      FROM      STUDENT      WHERE
Register_Number='$register_number'
    );
    if (mysqli_num_rows($check_student) == 0) {
        mysqli_query($conn, "
            INSERT INTO STUDENT
            (Student_Name, Register_Number, Course, Year, Institution)
            VALUES
            ('$student_name','$register_number','$course','$year','$institution')
        ");
        $student_id = mysqli_insert_id($conn);
    } else {
        $row = mysqli_fetch_assoc($check_student);
        $student_id = $row['Student_ID'];
    }
    $duplicate_check = mysqli_query($conn, "
    SELECT *
    FROM CERTIFICATE c
    JOIN STUDENT s ON c.Student_ID = s.Student_ID
    WHERE s.Register_Number = '$register_number'

```

```

AND c.Certificate_Type = '$certificate_type'
AND c.Status = 'Valid'
");
if (mysqli_num_rows($duplicate_check) > 0) {
    $message = "Certificate Already Exists For This Student";
    $showPopup = true;
} else {
    mysqli_query($conn, "
INSERT INTO CERTIFICATE
(Student_ID, Certificate_Type, Issue_Date, Certificate_File_Path, Status)
VALUES
('$student_id','$certificate_type',CURDATE(),',Valid')
");
    $auto_id = mysqli_insert_id($conn);
}
if (!isset($auto_id)) {
} else {
    if (isset($auto_id)) {
        $certificate_number = "CERT-" . date("Y") . "-" . str_pad($auto_id, 3, "0",
STR_PAD_LEFT);
        $pdf_path = "../certificates/" . $certificate_number . ".pdf";
        mysqli_query($conn, "
UPDATE CERTIFICATE
SET Certificate_File_Path='certificates/$certificate_number.pdf'
WHERE Certificate_ID='$auto_id'
");
        $verify_url = "http://localhost/digital_certificates/verify.php?id=" . $auto_id;
        $qr_api = "https://api.qrserver.com/v1/create-qr-code/?size=250x250&data="
        . urlencode($verify_url);
        $qr_path = "../qrcodes/" . $certificate_number . ".png";
        file_put_contents($qr_path, file_get_contents($qr_api));
        mysqli_query($conn, "

```

```

INSERT INTO QR_CODE (Certificate_ID, QR_Image, Verification_URL)
VALUES ('$auto_id','qrcores/$certificate_number.png','$verify_url')
");

$pdf = new Fpdi();
$pageCount = $pdf->setSourceFile("../templates/certificate_template.pdf");
$tpl = $pdf->importPage(1);
$size = $pdf->getTemplateSize($tpl);
$pdf->AddPage($size['orientation'], [$size['width'], $size['height']]);
$pdf->useTemplate($tpl, 0, 0, $size['width'], $size['height']);
$pdf->SetFont("Arial", "", 10);
$pdf->SetXY(25, 15);
$pdf->Cell(80, 8, "Certificate ID: " . $certificate_number, 0, 1, "L");
$pdf->SetFont("Times", "", 17);
$pdf->SetXY(80, 75);
$pdf->MultiCell(140, 6, "This certificate is proudly presented to ", 0, "C");
$pdf->SetFont("Courier", "B", 28);
$pdf->SetXY(80, 90);
$pdf->MultiCell(140, 6, $student_name, 0, "C");
$pdf->SetFont("Times", "", 17);
$pdf->SetXY(60, 105);
$text =
    "bearing Register Number " . $register_number .
    " for " . $certificate_type .
    " during the academic year " . $year .
    " in the Department of " . $course .
    " at " . $institution .
    ". This certificate is issued in recognition of the student's achievement " .
    "and its authenticity is subject to verification through the embedded QR code.";
$pdf->MultiCell(180, 6, $text, 0, "C");
$pdf->SetFont("Arial", "", 11);
$pdf->SetXY(30, 150);
$pdf->Cell(80, 8, "Date: " . date("d-m-Y"), 0, 1);
    
```

```

$pdf->Image($qr_path, 235, 140, 30);
$pdf->Link(235, 140, 30, 30, $verify_url);
$pdf->Output("F", $pdf_path);
$file_hash = hash_file("sha256", $pdf_path);
mysqli_query($conn, "
UPDATE CERTIFICATE
SET File_Hash='$file_hash'
WHERE Certificate_ID='$auto_id'
");
$download_link = "../certificates/" . $certificate_number . ".pdf";
$message = "Certificate Generated Successfully";
$showPopup = true;
}
}
}
?>
<!DOCTYPE html>
<html>
<head>
<style>
body {
background-image: url("1.png");
background-size: cover;
background-position: center;
background-repeat: no-repeat;
height: 100vh;
display: flex;
justify-content: center;
align-items: center;
}
.container {
background: rgb(250, 249, 253);

```

```
padding: 30px;
border-radius: 12px;
box-shadow: 0 0 12px gray;
text-align: center;
width: 450px;
}
input,
select {
width: 95%;
padding: 10px;
font-size: 15px;
border: 1px solid #ccc;
border-radius: 4px;
box-sizing: border-box;
}
select {
color: black;
}
.generate-btn {
width: 95%;
padding: 10px;
background-color: #2ecc71;
color: white;
border: none;
border-radius: 6px;
font-size: 16px;
cursor: pointer;
}
.revoke-btn {
width: 95%;
padding: 10px;
background-color: red;
```

```

        color: white;
        border: none;
        border-radius: 6px;
        font-size: 16px;
        cursor: pointer;
    }
    .popup {
        position: fixed;
        top: 50%;
        left: 50%;
        transform: translate(-50%, -50%);
        background: white;
        padding: 25px;
        border-radius: 10px;
        box-shadow: 0 0 15px gray;
        text-align: center;
        z-index: 999;
    }
</style>
</head>
<body>
    <?php if ($showPopup) { ?>
        <div class="popup">
            <h3><?php echo $message; ?></h3>
            <br>
            <?php if ($download_link != "") { ?>
                <a href="<?php echo $download_link; ?>" target="_blank" class="generate-
btn">
                    Download Certificate
                </a>
            <?php } ?>
        </div>
    
```



```

<?php } ?>
<div class="container">
    <h2>Certificate Management</h2>
    <form method="POST">
        Student Name:<br>
        <input type="text" name="student_name" placeholder="Enter Student
Name"><br><br>
        Register Number:<br>
        <input type="text" name="register_number" placeholder="Enter Register Number"
required><br><br>
        Course:<br>
        <input type="text" name="course" placeholder="Enter Course"><br><br>
        Academic Year:<br>
        <input type="text" name="year" placeholder="Enter Academic Year"><br><br>
        Institution:<br>
        <select name="institution" required
        style="width:95%; padding:10px; color:gray;"
        onchange="this.style.color='black'">
            <option value="" disabled selected hidden>Select Institution</option>
        <?php
        $inst = mysqli_query($conn, "SELECT * FROM INSTITUTION ORDER BY
Institution_Name ASC");
        while ($row = mysqli_fetch_assoc($inst)) {
            echo " <option value=" . $row['Institution_Name'] . ">" .
$row['Institution_Name'] . "</option>";
        }
        ?>
        </select><br><br>
        Certificate For:<br>
        <input type="text" name="certificate_type" placeholder="Enter Certificate
Type"><br><br>
        <button type="submit" name="generate" class="generate-btn">

```

Generate Certificate

</button>

<button type="submit" name="revoke" class="revoke-btn">

Revoke Certificate

</button>

</form>

</div>

</body>

</html>

CERTIFICATE MANAGEMENT MODULE:

manage_certificates.php:

```
<?php
include "../db.php";
$result = mysqli_query($conn, "
SELECT
    c.Certificate_ID,
    c.Certificate_Type,
    c.Issue_Date,
    c.Status,
    s.Student_Name,
    s.Register_Number
FROM CERTIFICATE c
JOIN STUDENT s
ON c.Student_ID = s.Student_ID
ORDER BY c.Certificate_ID DESC
");
?>
<!DOCTYPE html>
<html>
<head>
    <style>
        body {
            font-family: Arial;
            text-align: center;
            padding: 40px;
        }
        table {
            margin: auto;
            border-collapse: collapse;
            width: 90%;
```

```
}  
th,  
td {  
    border: 1px solid #ccc;  
    padding: 10px;  
}  
th {  
    background: #f2f2f2;  
}  
.valid {  
    color: green;  
    font-weight: bold;  
}  
.revoked {  
    color: red;  
    font-weight: bold;  
}  
button {  
    padding: 6px 10px;  
    border: none;  
    cursor: pointer;  
    border-radius: 4px;  
}  
.revoke-btn {  
    background: red;  
    color: white;  
}  
</style>  
</head>  
<body>  
    <h2>Manage Certificates</h2>  
    <table>
```

```

<tr>
    <th>ID</th>
    <th>Student Name</th>
    <th>Register No</th>
    <th>Type</th>
    <th>Issue Date</th>
    <th>Status</th>
    <th>Action</th>
</tr>
<?php while ($row = mysqli_fetch_assoc($result)) { ?>
    <tr>
        <td><?php echo $row['Certificate_ID']; ?></td>
        <td><?php echo $row['Student_Name']; ?></td>
        <td><?php echo $row['Register_Number']; ?></td>
        <td><?php echo $row['Certificate_Type']; ?></td>
        <td><?php echo $row['Issue_Date']; ?></td>
        <td class="<?php echo strtolower($row['Status']); ?>">
            <?php echo $row['Status']; ?>
        </td>
        <td>
            <?php if ($row['Status'] == 'Valid') { ?>
                <a href="revoke_certificate.php?id=<?php echo $row['Certificate_ID'];
?>">
                    <button class="revoke-btn">Revoke</button>
                </a>
            <?php } else { ?>
                <?php } ?>
            </td>
        </tr>
    <?php } ?>
</table>
<br><br>

```

```
<a href="generate_certificate.php">Back to Generate</a>
</body>
</html>
```

CERTIFICATE REVOCATION MODULE:

revoke_certificate.php:

```
<?php
include "../db.php";
if (!isset($_GET['id'])) {
    echo "Invalid Request";
    exit;
}
$certificate_id = $_GET['id'];
mysqli_query($conn, "
UPDATE CERTIFICATE
SET Status = 'Revoked'
WHERE Certificate_ID = '$certificate_id'
");
echo "<h3>Certificate Revoked Successfully</h3>";
echo "<a href='generate_certificate.php'>Back</a>";
?>
```

EXTERNAL CERTIFICATE UPLOAD MODULE:

upload_external.php:

```
<?php
error_reporting(E_ALL);
ini_set('display_errors', 1);
include "../db.php";
require_once __DIR__ . '/../fpdf/fpdf.php';
require_once __DIR__ . '/../fpdi/src/autoload.php';
use setasign\Fpdi\Fpdi;
$message = "";
if (isset($_POST['upload'])) {
    $get_student = mysqli_query($conn, "
SELECT Student_ID
FROM STUDENT
WHERE Student_Name='External Student'
LIMIT 1
");
    if (mysqli_num_rows($get_student) == 0) {
        mysqli_query($conn, "
INSERT INTO STUDENT
(Student_Name, Register_Number, Course, Year, Institution)
VALUES
('External Student','EXT" . time() . "','External','External','External')
");
        $student_id = mysqli_insert_id($conn);
    } else {
        $row = mysqli_fetch_assoc($get_student);
        $student_id = $row['Student_ID'];
    }
    $certificate_number = "EXT-" . date("Y") . "-" . time();
    $pdf_path = "../certificates/" . $certificate_number . ".pdf";
    mysqli_query($conn, "
```

INSERT INTO CERTIFICATE

(Student_ID, Certificate_Type, Issue_Date, Certificate_File_Path, Status)

VALUES

('Student_id','External',CURDATE(),'certificates/\$certificate_number.pdf','Valid')

");

\$auto_id = mysqli_insert_id(\$conn);

\$verify_url = "http://localhost/digital_certificates/verify.php?id=" . \$auto_id;

\$qr_api = "https://api.qrserver.com/v1/create-qr-code/?size=250x250&data="

. urlencode(\$verify_url);

\$qr_path = "../qrcodes/" . \$certificate_number . ".png";

file_put_contents(\$qr_path, file_get_contents(\$qr_api));

mysqli_query(\$conn, "

INSERT INTO QR_CODE

(Certificate_ID, QR_Image, Verification_URL)

VALUES

('Student_id','qrcodes/\$certificate_number.png','\$verify_url')

");

\$tmp_pdf = \$_FILES['external_pdf']['tmp_name'];

\$pdf = new Fpdf();

\$pageCount = \$pdf->setSourceFile(\$tmp_pdf);

\$tpl = \$pdf->importPage(1);

\$size = \$pdf->getTemplateSize(\$tpl);

\$pdf->AddPage(\$size['orientation'], [\$size['width'], \$size['height']]);

\$pdf->useTemplate(\$tpl, 0, 0, \$size['width'], \$size['height']);

\$pdf->Image(\$qr_path, 235, 140, 30);

\$pdf->Output("F", \$pdf_path);

\$file_hash = hash_file("sha256", \$pdf_path);

mysqli_query(\$conn, "

UPDATE CERTIFICATE

SET File_Hash='\$file_hash'

WHERE Certificate_ID='\$auto_id'

");


```
$message = "External Certificate Uploaded Successfully";  
}  
?>  
<!DOCTYPE html>  
<html>  
<head>  
  <style>  
    body {  
      background-image: url("1.png");  
      background-size: cover;  
      background-position: center;  
      background-repeat: no-repeat;  
      font-family: Arial;  
      text-align: center;  
      padding-top: 80px;  
    }  
    .container {  
      display: inline-block;  
      background: white;  
      padding: 30px;  
      border-radius: 10px;  
      box-shadow: 0 0 12px gray;  
      width: 400px;  
    }  
    input {  
      width: 90%;  
      padding: 10px;  
    }  
    button {  
      padding: 10px 20px;  
      background: #2ecc71;  
      color: white;
```

```
        border: none;
        border-radius: 6px;
        cursor: pointer;
    }
</style>
</head>
<body>
    <div class="container">
        <h2>Upload External Certificate</h2>
        <form method="POST" enctype="multipart/form-data">
            <input type="file" name="external_pdf" accept="application/pdf" required>
            <br><br>
            <button type="submit" name="upload">Upload Certificate</button>
        </form>
        <br>
        <?php
            if ($message != "") {
                echo "<b>" . $message . "</b>";
            }
        ?>
    </div>
</body>
</html>
```

VERIFICATION LOG MODULE:

view_verification_logs.php:

```
<?php
include "../db.php";
$result = mysqli_query($conn, "
SELECT
    v.Verification_ID,
    v.Certificate_ID,
    v.Scan_Date,
    v.Verification_Result,
    s.Student_Name,
    s.Register_Number
FROM VERIFICATION v
JOIN CERTIFICATE c ON v.Certificate_ID = c.Certificate_ID
JOIN STUDENT s ON c.Student_ID = s.Student_ID
ORDER BY v.Scan_Date DESC
");
?>
<!DOCTYPE html>
<html>
<head>
    <title>Verification Logs</title>
    <style>
        body {
            background-image: url("1.png");
            background-size: cover;
            background-position: center;
            background-repeat: no-repeat;
            font-family: Arial;
            text-align: center;
            padding-top: 40px;
```

```

    }
    table {
        background-color: white;
        border-collapse: collapse;
        margin: auto;
        width: 80%;
    }
    th,
    td {
        border: 1px solid #ccc;
        padding: 8px;
    }
    th {
        background: #2ecc71;
        color: white;
    }
</style>
</head>
<body>
    <h2>Verification Logs</h2>
    <table>
        <tr>
            <th>Certificate ID</th>
            <th>Student Name</th>
            <th>Register Number</th>
            <th>Scan Date</th>
            <th>Result</th>
        </tr>
        <?php while ($row = mysqli_fetch_assoc($result)) { ?>
            <tr>
                <td><?php echo $row['Certificate_ID']; ?></td>
                <td><?php echo $row['Student_Name']; ?></td>

```

```
<td><?php echo $row['Register_Number']; ?></td>  
<td><?php echo $row['Scan_Date']; ?></td>  
<td><?php echo $row['Verification_Result']; ?></td>  
</tr>  
<?php } ?>  
</table>  
</body>  
</html>
```

STUDENT VERIFICATION MODULE:

student_verify.php:

```
<?php
include "db.php";
$showResult = false;
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $student_name = strtoupper($_POST['student_name']);
    $register_number = $_POST['register_number'];
    $student_query = mysqli_query($conn, "
    SELECT Student_ID, Student_Name
    FROM STUDENT
    WHERE UPPER(Student_Name) = '$student_name'
    AND Register_Number = '$register_number'
    ");
    if (mysqli_num_rows($student_query) > 0) {
        $student_data = mysqli_fetch_assoc($student_query);
        $student_id = $student_data['Student_ID'];
        $cert_query = mysqli_query($conn, "
        SELECT *
        FROM CERTIFICATE
        WHERE Student_ID = '$student_id'
        ORDER BY Certificate_ID DESC
        ");
        $showResult = true;
    } else {
        $error = "Invalid Details";
    }
}
?>
<!DOCTYPE html>
<html>
<head>
```

<style>

```
body {  
    background-image: url("admin/1.png");  
    background-size: cover;  
    background-position: center;  
    background-repeat: no-repeat;  
    font-family: Arial;  
    text-align: center;  
    padding-top: 40px;  
}
```

```
.container {  
    display: inline-block;  
    padding: 25px;  
    border-radius: 10px;  
    box-shadow: 0 0 12px gray;  
    width: 450px;  
    background: white;  
}
```

```
input {  
    width: 90%;  
    padding: 8px;  
    margin: 5px;  
}
```

```
button {  
    padding: 8px 15px;  
    background: #2ecc71;  
    color: white;  
    border: none;  
    border-radius: 5px;  
    cursor: pointer;  
}
```

```
.valid {
```

```

        color: green;
    }
    .revoked {
        color: red;
    }
    .tampered {
        color: orange;
    }
    .cert-box {
        margin-top: 15px;
        padding: 10px;
        border-top: 1px solid #ccc;
        word-wrap: break-word;
    }
</style>
</head>
<body>
    <div class="container">
        <h2>Student Certificate Verification</h2>
        <form method="POST">
            Student Name:<br>
            <input type="text" name="student_name" required><br>
            Register Number:<br>
            <input type="text" name="register_number" required><br><br>
            <button type="submit">Verify</button>
        </form>
        <?php if (isset($error)) { ?>
            <p style="color:red;"><?php echo $error; ?></p>
        <?php } ?>
        <?php if ($showResult) { ?>

            <hr>

```



```

<h3>Certificate Details</h3>
<?php
$valid_count = 0;
$revoked_count = 0;
$stampered_count = 0;
$total = 0;
while ($data = mysqli_fetch_assoc($cert_query)) {
    $pdf_path = $data['Certificate_File_Path'];
    if (file_exists($pdf_path)) {
        $current_hash = hash_file("sha256", $pdf_path);
        if ($current_hash != $data['File_Hash']) {
            $data['Status'] = "Tampered";
        }
    }
    $total++;
    if ($data['Status'] == "Valid") {
        $valid_count++;
    } else if ($data['Status'] == "Revoked") {
        $revoked_count++;
    } else if ($data['Status'] == "Tampered") {
        $stampered_count++;
    }
}
?>
<div class="cert-box">
    <b>Certificate ID:</b> <?php echo $data['Certificate_ID']; ?><br>
    <div style="word-wrap:break-word;">
        <b>Certificate Type:</b> <?php echo $data['Certificate_Type']; ?>
    </div>
    <b>Issue Date:</b> <?php echo $data['Issue_Date']; ?><br>
    <b>Status:</b>
    <span class="<?php echo strtolower($data['Status']); ?>">
        <?php echo $data['Status']; ?>
    </span>

```

```

</span>
<br><br>
<?php if ($data['Status'] == 'Valid') { ?>
    <a href="<?php echo $data['Certificate_File_Path']; ?>" target="_blank">
        Download Certificate
    </a>
<?php } elseif ($data['Status'] == 'Revoked') { ?>
    <p      class="revoked"><b>Download      Disabled      (Certificate
Revoked)</b></p>
    <?php } elseif ($data['Status'] == 'Tampered') { ?>
    <p      class="tampered"><b>Certificate      Tampered      -      Download
Disabled</b></p>
    <?php } ?>
</div>
<?php } ?>
<hr>
<h3>Certificate Summary</h3>
<p><b>Total Certificates:</b> <?php echo $total; ?></p>
<p class="valid"><b>Valid Certificates:</b> <?php echo $valid_count; ?></p>
<p class="revoked"><b>Revoked Certificates:</b> <?php echo $revoked_count;
?></p>
    <p      class="tampered"><b>Tampered      Certificates:</b>      <?php      echo
$stampered_count; ?></p>
    <?php } ?>
</div>
</body>
</html>

```

PUBLIC CERTIFICATE VERIFICATION MODULE:

verify.php:

```
<?php
include "db.php";
if (!isset($_GET['id'])) {
    echo "<h2>Invalid Request</h2>";
    exit;
}
$certificate_id = $_GET['id'];
$check = mysqli_query($conn, "
SELECT
    c.Certificate_ID,
    c.Certificate_Type,
    c.Issue_Date,
    c.Status,
    c.Certificate_File_Path,
    c.File_Hash,
    s.Student_Name,
    s.Register_Number,
    s.Course,
    s.Year,
    s.Institution
FROM CERTIFICATE c
JOIN STUDENT s
ON c.Student_ID = s.Student_ID
WHERE c.Certificate_ID = '$certificate_id'
");
if (mysqli_num_rows($check) == 0) {
    mysqli_query($conn, "
INSERT INTO VERIFICATION
(Certificate_ID, Scan_Date, Verification_Result)
```

```
VALUES ('$certificate_id', NOW(), 'Invalid')
");
echo "<h2>Invalid Certificate</h2>";
exit;
}
$row = mysqli_fetch_assoc($check);
$file_path = $row['Certificate_File_Path'];
if (!file_exists($file_path)) {
    $file_path = "certificates/" . $file_path;
}
$current_hash = hash_file("sha256", $file_path);
if ($current_hash != $row['File_Hash']) {
    mysqli_query($conn, "
INSERT INTO VERIFICATION
(Certificate_ID, Scan_Date, Verification_Result)
VALUES ('$certificate_id', NOW(), 'Tampered')
");
    echo "<h2>Certificate Tampered</h2>";
    exit;
}
if ($row['Status'] != 'Valid') {
    mysqli_query($conn, "
INSERT INTO VERIFICATION
(Certificate_ID, Scan_Date, Verification_Result)
VALUES ('$certificate_id', NOW(), 'Invalid')
");
    echo "<h2>Invalid Certificate</h2>";
    exit;
}
mysqli_query($conn, "
INSERT INTO VERIFICATION
(Certificate_ID, Scan_Date, Verification_Result)
```

```
VALUES ('$certificate_id', NOW(), 'Valid')
");
$reg = $row['Register_Number'];
if (strlen($reg) > 5) {
    $masked_reg = substr($reg, 0, 3) .
        str_repeat("*", strlen($reg) - 5) .
        substr($reg, -2);
} else {
    $masked_reg = $reg;
}
$name_parts = explode(" ", $row['Student_Name']);
$masked_name = "";
foreach ($name_parts as $part) {
    if (strlen($part) > 2) {
        $masked_name .= substr($part, 0, 2) .
            str_repeat("*", strlen($part) - 2) . " ";
    } else {
        $masked_name .= $part . " ";
    }
}
$masked_name = trim($masked_name);
?>
<!DOCTYPE html>
<html>
<head>
    <title>Certificate Verification</title>
    <style>
        body {
            font-family: Arial;
            text-align: center;
            padding-top: 50px;
            background: #f4f4f4;
```

```

    }
    .container {
        display: inline-block;
        background: white;
        padding: 25px;
        border-radius: 10px;
        box-shadow: 0 0 12px gray;
        width: 420px;
    }
    .valid {
        color: green;
    }
</style>
</head>
<body>
    <div class="container">
        <h2 class="valid">Certificate Verified</h2>
        <?php if ($row['Certificate_Type'] != 'External') { ?>
            <b>Student Name:</b> <?php echo $masked_name; ?><br>
            <b>Register Number:</b> <?php echo $masked_reg; ?><br>
            <b>Course:</b> <?php echo $row['Course']; ?><br>
            <b>Year:</b> <?php echo $row['Year']; ?><br>
            <b>Institution:</b> <?php echo $row['Institution']; ?><br><br>
        <?php } ?>
        <b>Certificate Type:</b> <?php echo $row['Certificate_Type']; ?><br>
        <b>Issue Date:</b> <?php echo $row['Issue_Date']; ?><br>
        <b>Status:</b> <?php echo $row['Status']; ?><br><br>
        <a href="<?php echo $row['Certificate_File_Path']; ?>" target="_blank">
            View / Download Certificate
        </a>
    </div>
</body></html>

```

PDF GENERATION TESTING MODULE:

fpdf_test.php:

```
<?php  
require("fpdf/fpdf.php");  
$pdf = new FPDF();  
$pdf->AddPage();  
$pdf->SetFont("Arial", "B", 16);  
$pdf->Cell(0, 10, "FPDF Working", 0, 1, "C");  
$pdf->Output();  
?>
```

7.3 SQL DATABASE:

CREATE DATABASE:

```
CREATE DATABASE digital_certificate_db;  
USE digital_certificate_db;
```

CREATE ADMIN TABLE:

```
CREATE TABLE ADMIN (  
Admin_ID INT AUTO_INCREMENT PRIMARY KEY,  
Username VARCHAR(50) NOT NULL,  
Password VARCHAR(100) NOT NULL  
);
```

Insert admin login:

```
INSERT INTO ADMIN (Username, Password)  
VALUES ('admin','admin123');
```

CREATE STUDENT TABLE:

```
CREATE TABLE STUDENT (  
Student_ID INT AUTO_INCREMENT PRIMARY KEY,  
Student_Name VARCHAR(100),  
Register_Number VARCHAR(50) UNIQUE,  
Course VARCHAR(100),  
Year VARCHAR(20),  
Institution VARCHAR(150)  
);
```


CREATE CERTIFICATE TABLE:

```
CREATE TABLE CERTIFICATE (  
Certificate_ID INT AUTO_INCREMENT PRIMARY KEY,  
Student_ID INT,  
Certificate_Type VARCHAR(200),  
Issue_Date DATE,  
Certificate_File_Path VARCHAR(255),  
Status VARCHAR(20),  
File_Hash VARCHAR(256),  
FOREIGN KEY (Student_ID) REFERENCES STUDENT(Student_ID)  
ON DELETE CASCADE  
);
```

CREATE QR_CODE TABLE:

```
CREATE TABLE QR_CODE (  
QR_ID INT AUTO_INCREMENT PRIMARY KEY,  
Certificate_ID INT,  
QR_Image VARCHAR(255),  
Verification_URL VARCHAR(255),  
FOREIGN KEY (Certificate_ID) REFERENCES CERTIFICATE(Certificate_ID)  
ON DELETE CASCADE  
);
```

CREATE VERIFICATION TABLE:

```
CREATE TABLE VERIFICATION (  
Verification_ID INT AUTO_INCREMENT PRIMARY KEY,  
Certificate_ID INT,  
Scan_Date DATETIME,  
Verification_Result VARCHAR(50),  
FOREIGN KEY (Certificate_ID) REFERENCES CERTIFICATE(Certificate_ID)  
ON DELETE CASCADE  
);
```

CREATE INSTITUTION TABLE:

```
CREATE TABLE INSTITUTION (  
Institution_ID INT AUTO_INCREMENT PRIMARY KEY,  
Institution_Name VARCHAR(150)  
);
```

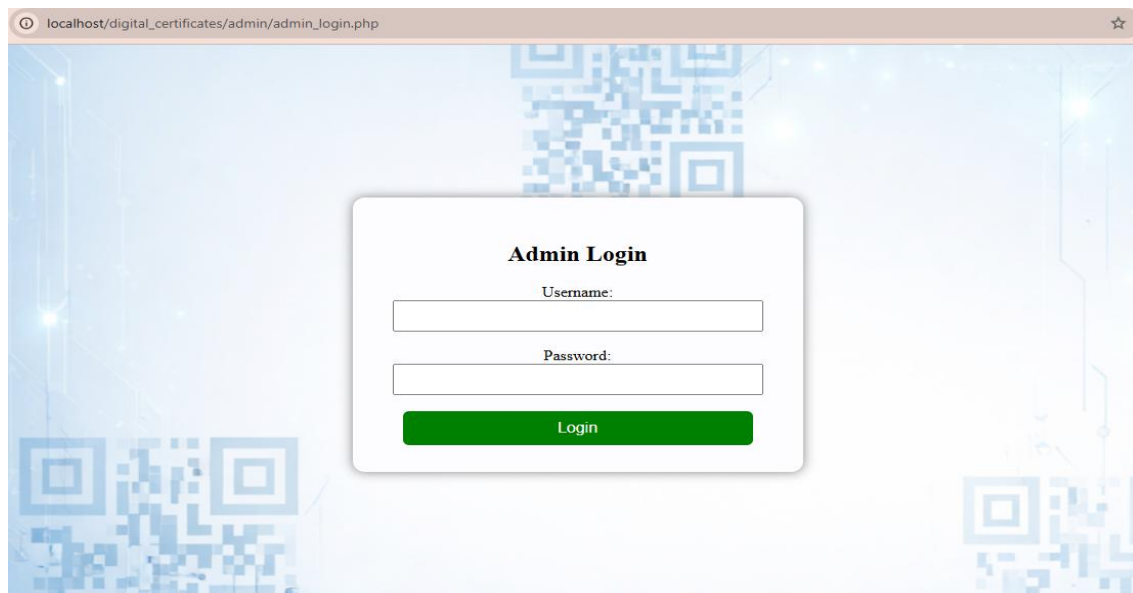
INSERT INSTITUTIONS:

```
INSERT INTO INSTITUTION (Institution_Name) VALUES  
( 'Presidency College'),  
( 'Christ University'),  
( 'Jain University'),  
( 'Bangalore University')
```

SCREENSHOTS

8. SCREENSHOTS:

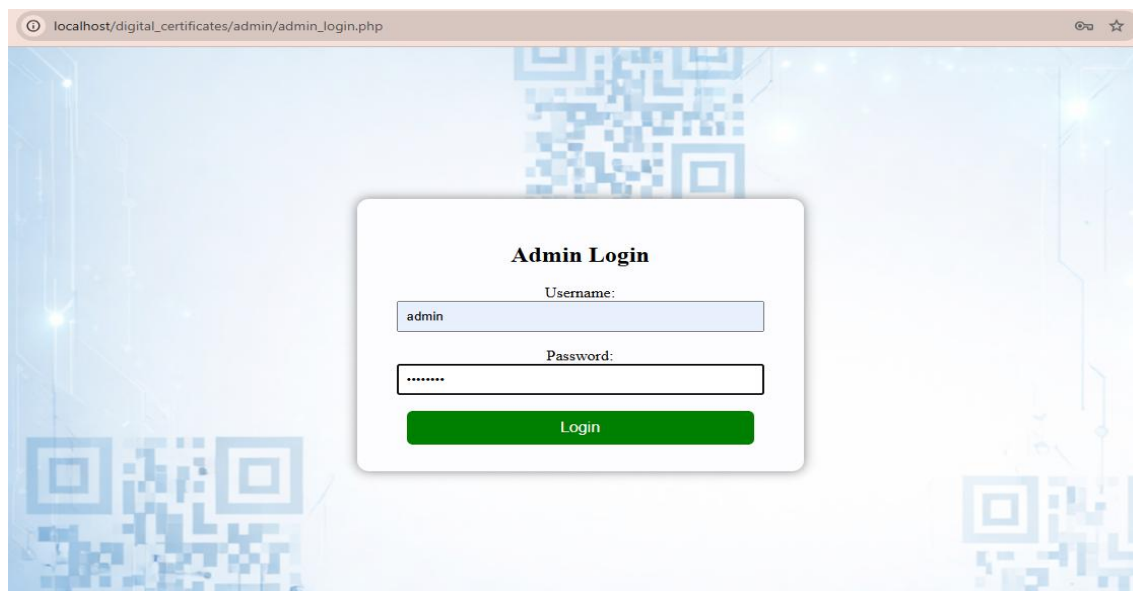
ADMIN LOGIN PAGE:



The screenshot shows a web browser window with the address bar displaying 'localhost/digital_certificates/admin/admin_login.php'. The page has a light blue background with a subtle QR code pattern. In the center, there is a white box titled 'Admin Login'. Inside this box, there are two input fields: 'Username:' and 'Password:'. Below these fields is a green button labeled 'Login'.

Fig 8.1

ADMIN LOGIN WITH ENTERED CREDENTIALS:



The screenshot shows the same 'Admin Login' page as Fig 8.1, but with the 'Username' field filled with 'admin' and the 'Password' field filled with '*****'. The green 'Login' button remains visible below the fields.

Fig 8.2

ADMIN LOGIN SUCCESSFUL PAGE:

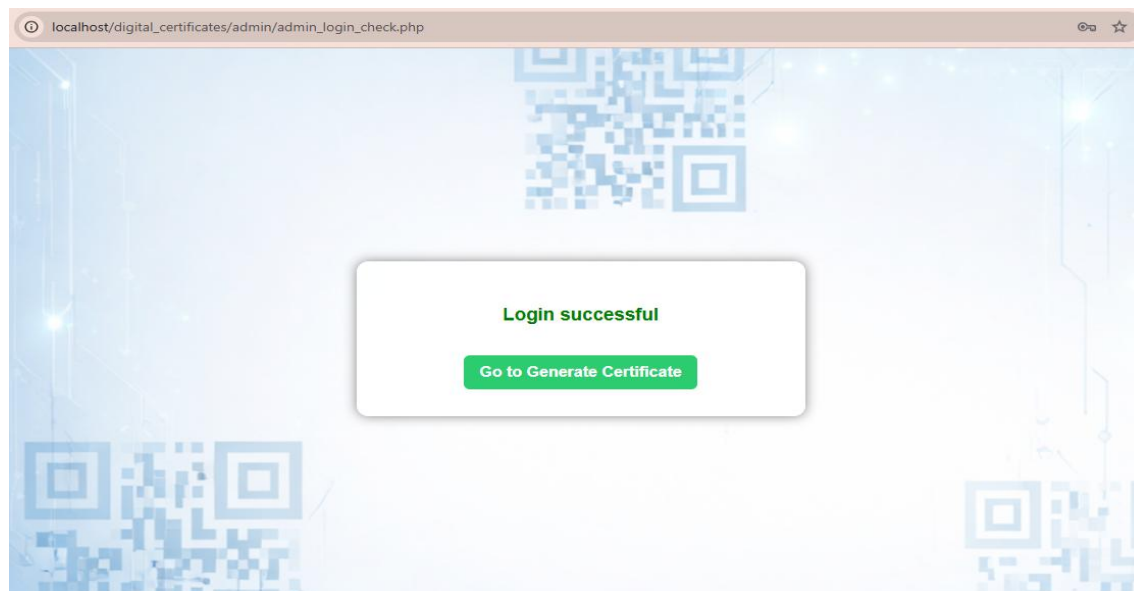


Fig 8.3

ADMIN HOME PAGE:

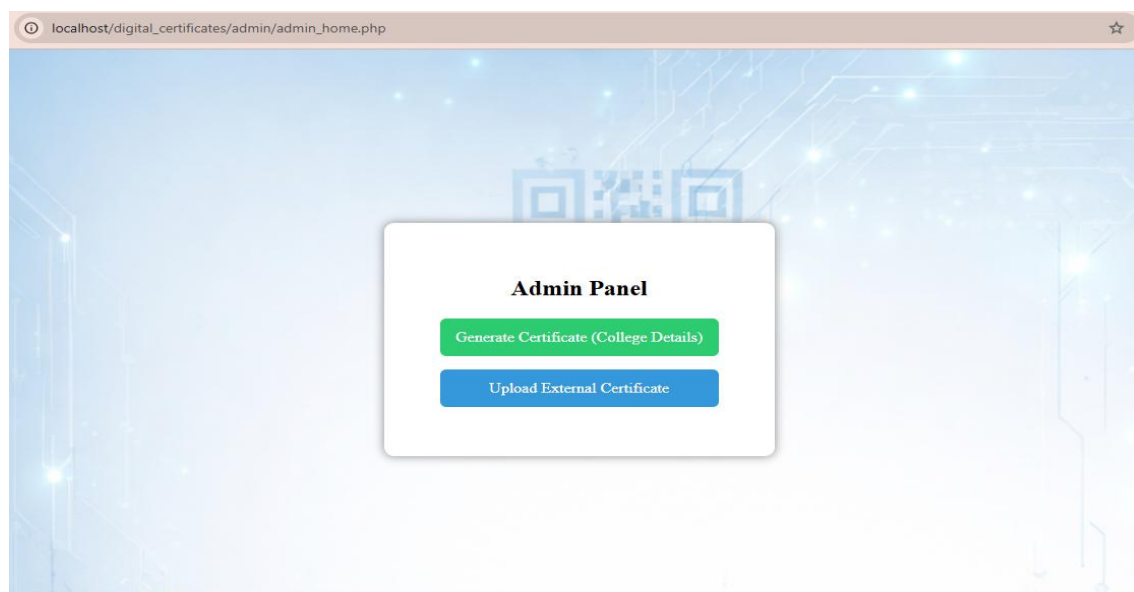
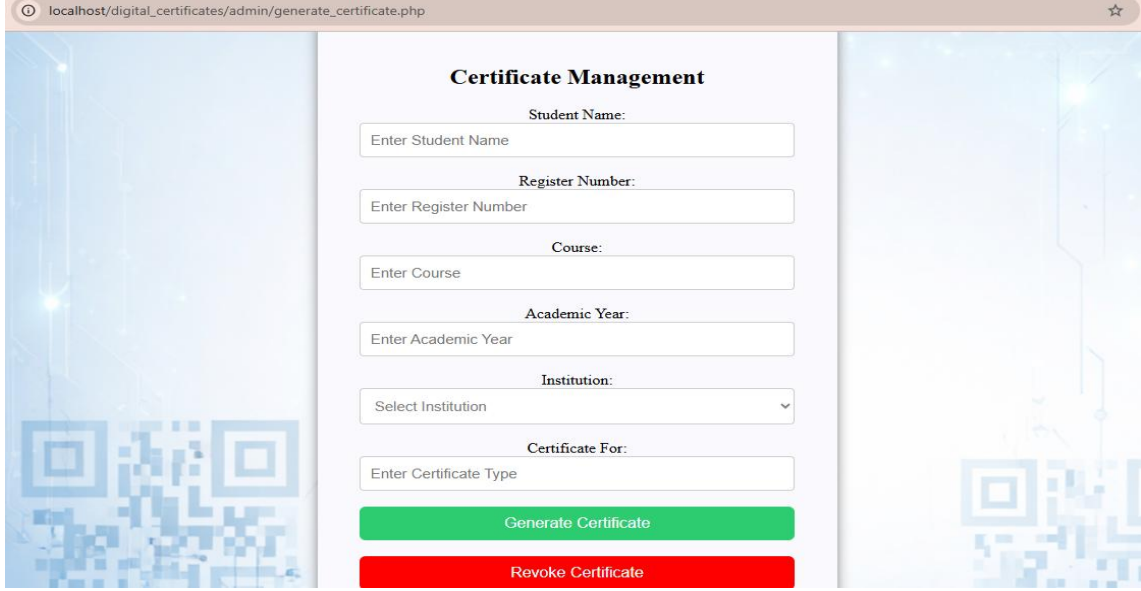


Fig 8.4

CERTIFICATE GENERATION PAGE:

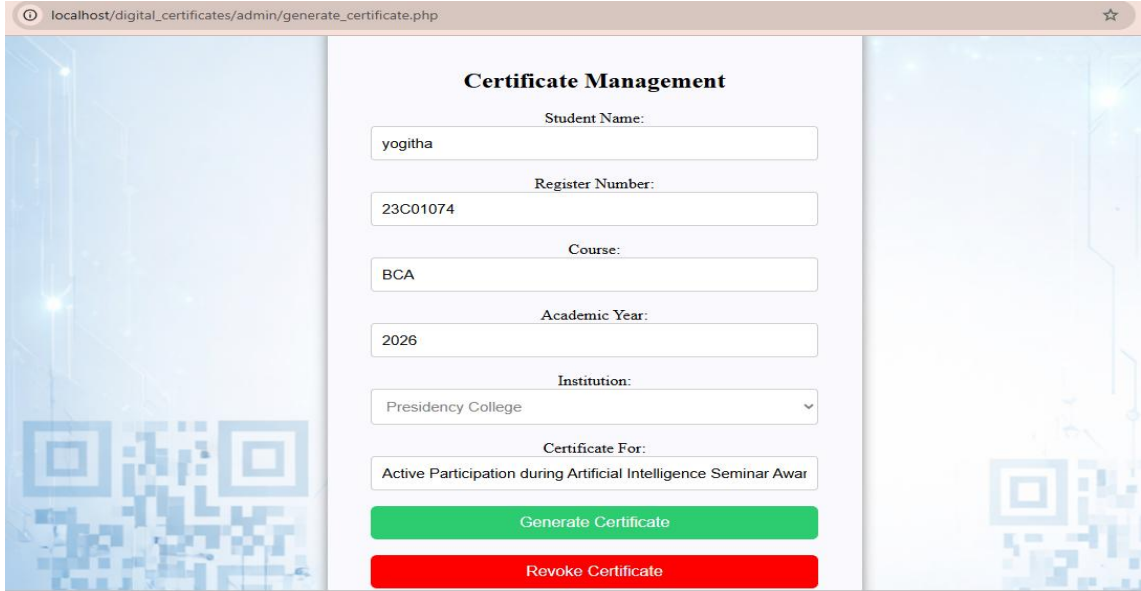


The screenshot shows a web browser window with the address bar displaying 'localhost/digital_certificates/admin/generate_certificate.php'. The page title is 'Certificate Management'. The form contains the following fields and buttons:

- Student Name:
- Register Number:
- Course:
- Academic Year:
- Institution:
- Certificate For:
-
-

Fig 8.5

CERTIFICATE GENERATION WITH FILLED DETAILS:

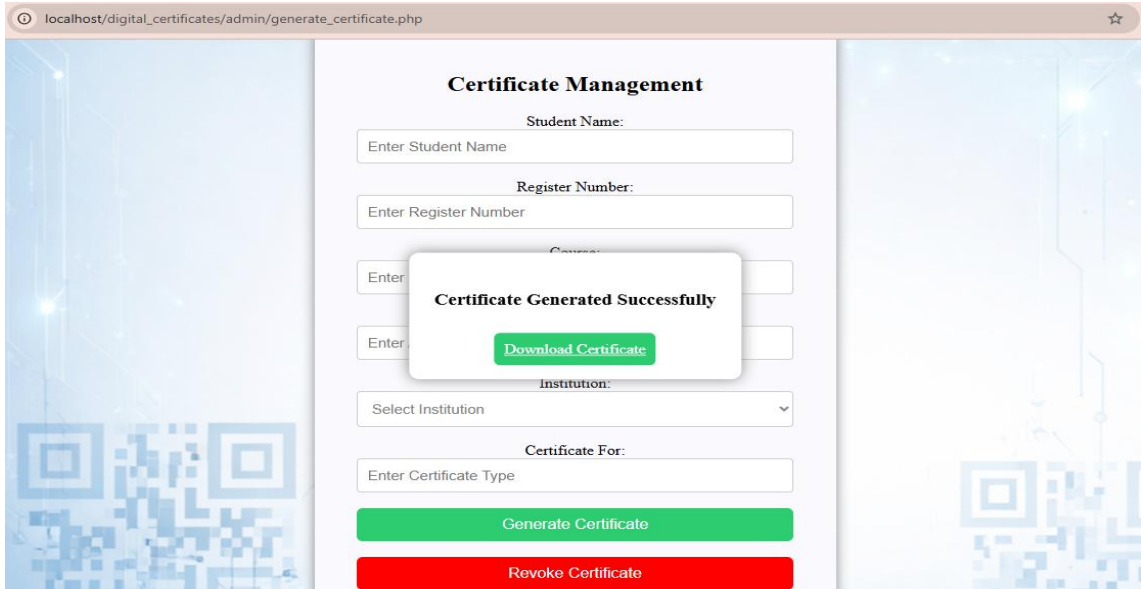


The screenshot shows the same web browser window as Fig 8.5, but with the following details filled in:

- Student Name:
- Register Number:
- Course:
- Academic Year:
- Institution:
- Certificate For:
-
-

Fig 8.6

CERTIFICATE GENERATED SUCCESSFULLY POPUP:



The screenshot shows a web browser window with the URL `localhost/digital_certificates/admin/generate_certificate.php`. The page is titled "Certificate Management" and contains several input fields for "Student Name", "Register Number", "Gender", "Institution", and "Certificate For". A green "Generate Certificate" button is visible. A modal popup is displayed in the center with the text "Certificate Generated Successfully" and a green "Download Certificate" button. The background of the page features a blue and white geometric pattern with QR codes.

Fig 8.7

GENERATED CERTIFICATE PDF WITH QR CODE:

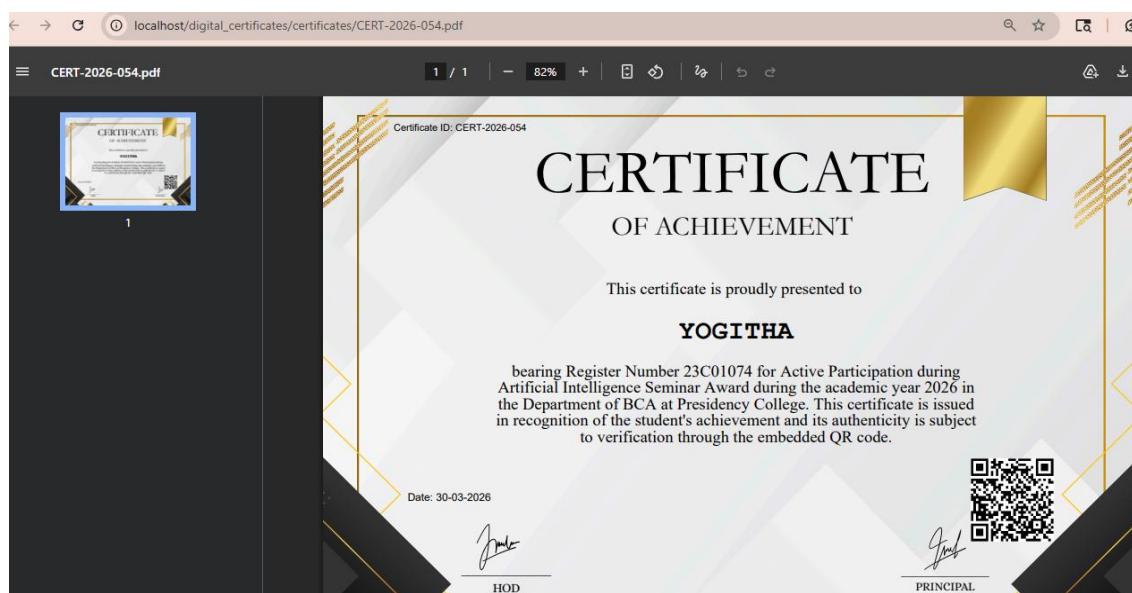


Fig 8.8

QR CODE VERIFICATION RESULT:

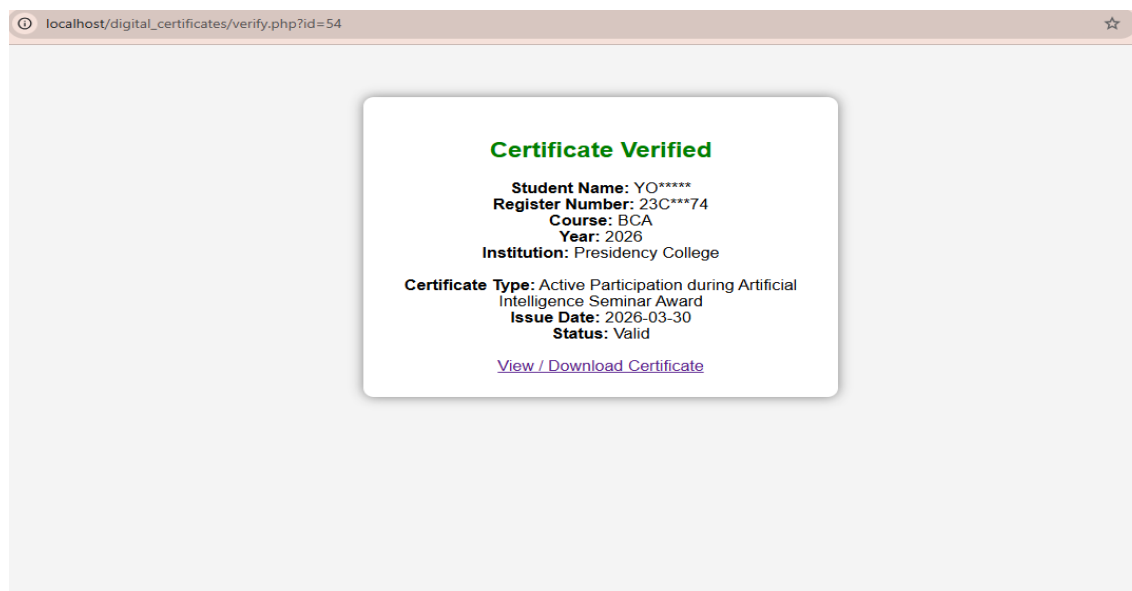


Fig 8.9

UPLOAD EXTERNAL CERTIFICATE PAGE:

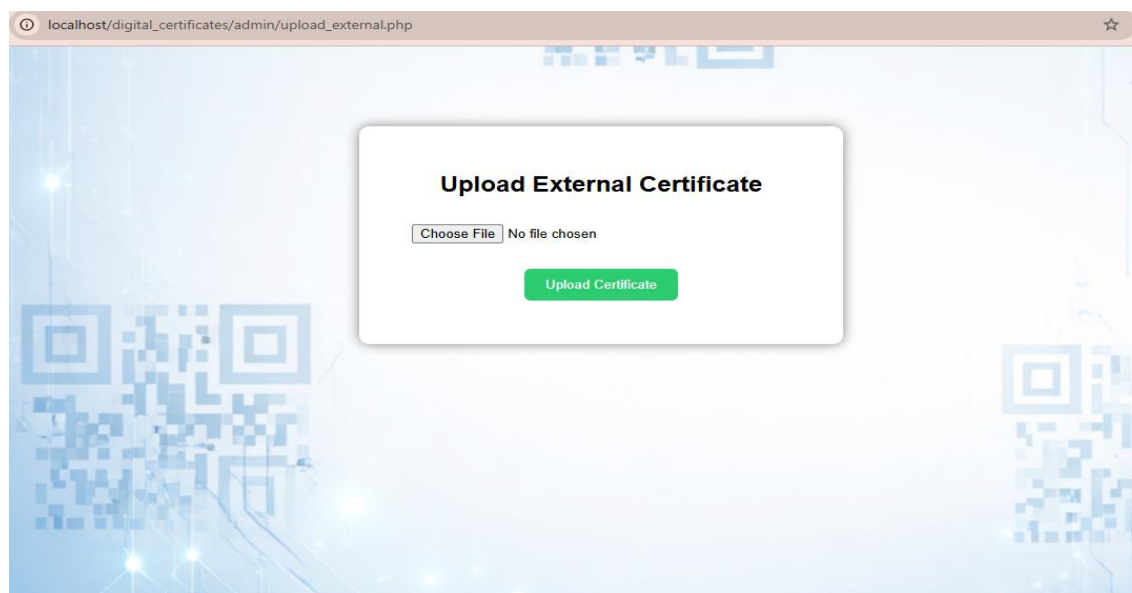


Fig 8.10

EXTERNAL CERTIFICATE FILE SELECTED:

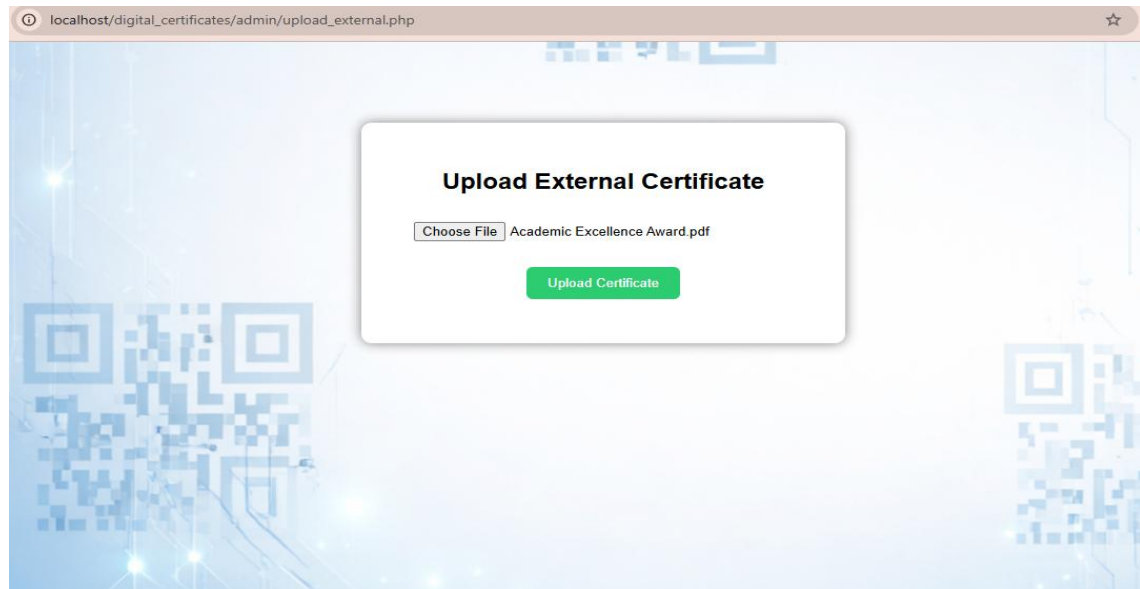


Fig 8.11

EXTERNAL CERTIFICATE UPLOADED SUCCESSFULLY:

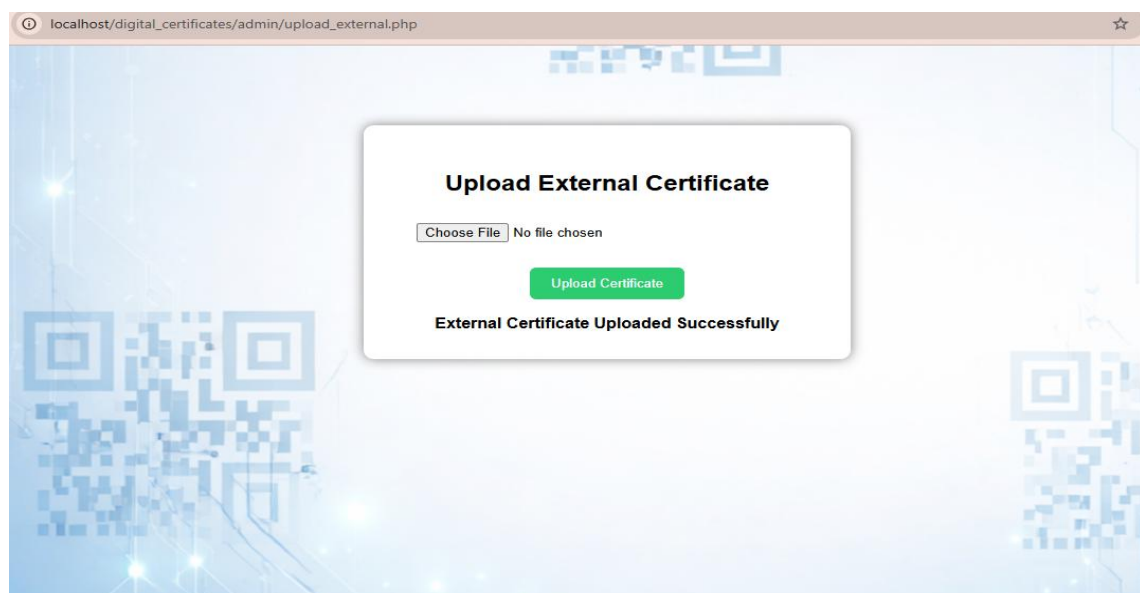


Fig 8.12

GENERATED EXTERNAL CERTIFICATE WITH QR CODE:



Fig 8.13

EXTERNAL CERTIFICATE VERIFICATION RESULT:

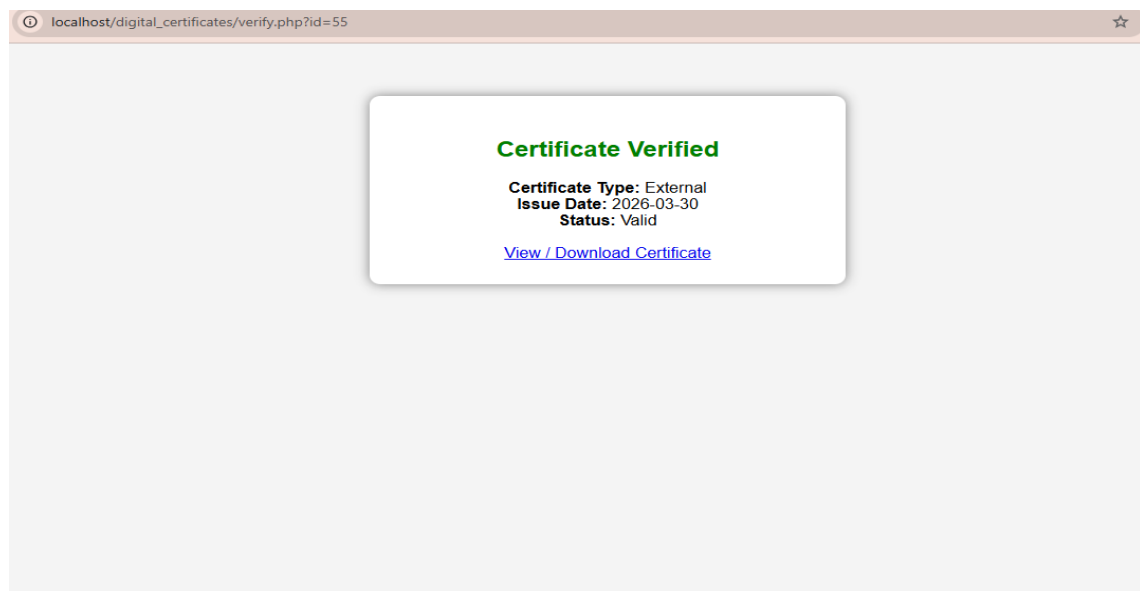


Fig 8.14

ADMIN DASHBOARD:

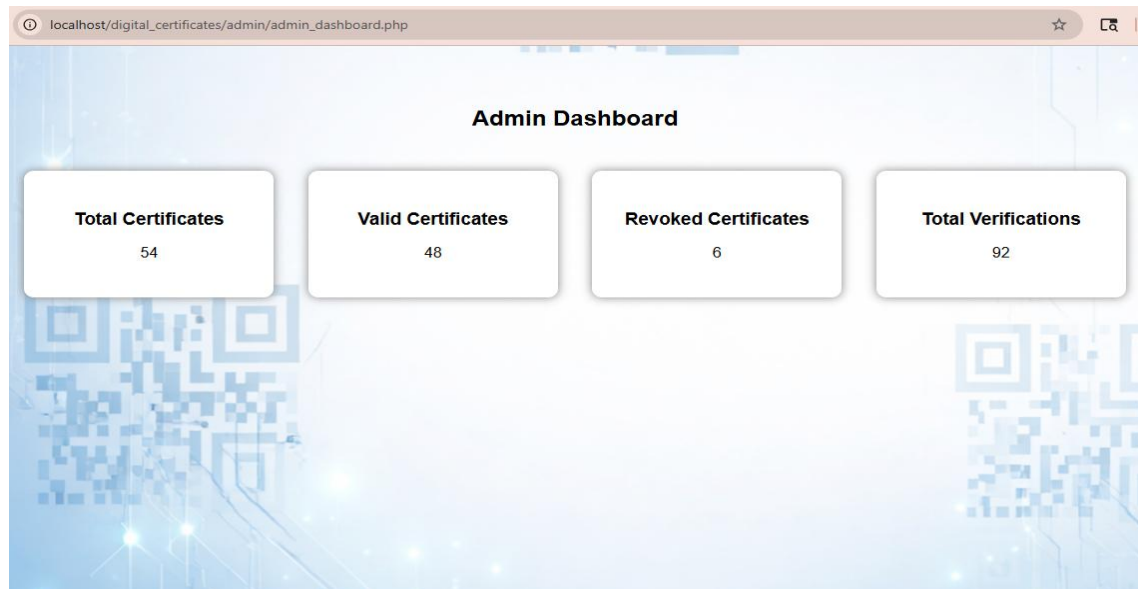
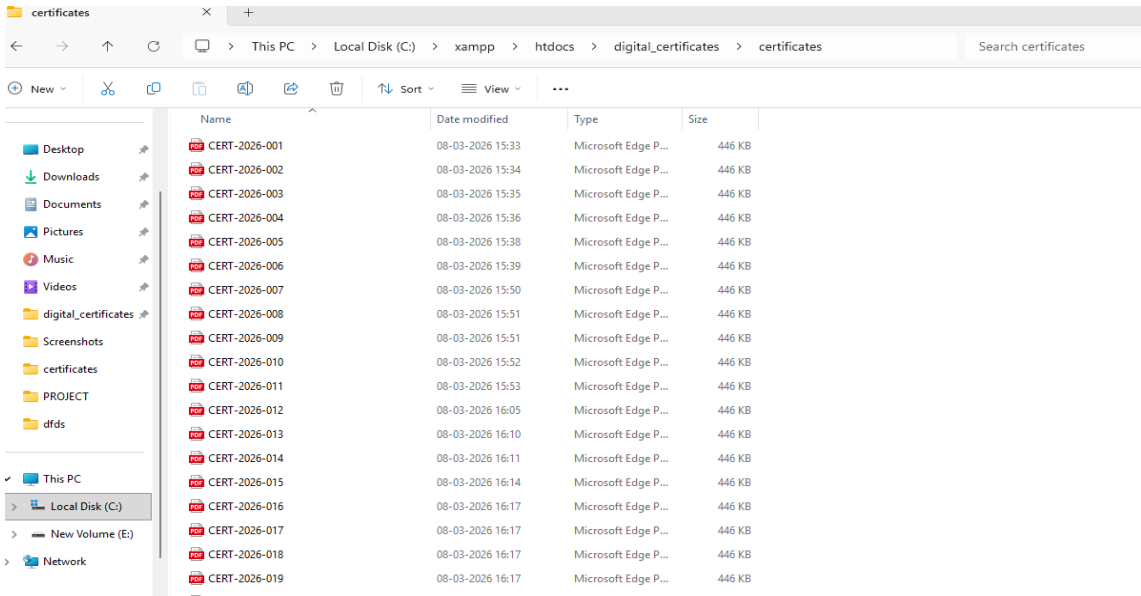


Fig 8.15

CERTIFICATES STORED IN SYSTEM FOLDER:



The screenshot shows a Windows File Explorer window with the address bar displaying the path: `This PC > Local Disk (C:) > xampp > htdocs > digital_certificates > certificates`. The left sidebar shows the folder structure, and the main pane displays a list of certificate files.

Name	Date modified	Type	Size
CERT-2026-001	08-03-2026 15:33	Microsoft Edge P...	446 KB
CERT-2026-002	08-03-2026 15:34	Microsoft Edge P...	446 KB
CERT-2026-003	08-03-2026 15:35	Microsoft Edge P...	446 KB
CERT-2026-004	08-03-2026 15:36	Microsoft Edge P...	446 KB
CERT-2026-005	08-03-2026 15:38	Microsoft Edge P...	446 KB
CERT-2026-006	08-03-2026 15:39	Microsoft Edge P...	446 KB
CERT-2026-007	08-03-2026 15:50	Microsoft Edge P...	446 KB
CERT-2026-008	08-03-2026 15:51	Microsoft Edge P...	446 KB
CERT-2026-009	08-03-2026 15:51	Microsoft Edge P...	446 KB
CERT-2026-010	08-03-2026 15:52	Microsoft Edge P...	446 KB
CERT-2026-011	08-03-2026 15:53	Microsoft Edge P...	446 KB
CERT-2026-012	08-03-2026 16:05	Microsoft Edge P...	446 KB
CERT-2026-013	08-03-2026 16:10	Microsoft Edge P...	446 KB
CERT-2026-014	08-03-2026 16:11	Microsoft Edge P...	446 KB
CERT-2026-015	08-03-2026 16:14	Microsoft Edge P...	446 KB
CERT-2026-016	08-03-2026 16:17	Microsoft Edge P...	446 KB
CERT-2026-017	08-03-2026 16:17	Microsoft Edge P...	446 KB
CERT-2026-018	08-03-2026 16:17	Microsoft Edge P...	446 KB
CERT-2026-019	08-03-2026 16:17	Microsoft Edge P...	446 KB

Fig 8.16

VERIFICATION LOGS PAGE:

localhost/digital_certificates/admin/view_verification_logs.php

Verification Logs				
Certificate ID	Student Name	Register Number	Scan Date	Result
55	External Student	EXT1772977565	2026-03-30 20:05:32	Valid
54	YOGITHA	23C01074	2026-03-30 20:01:43	Valid
54	YOGITHA	23C01074	2026-03-30 20:01:22	Valid
54	YOGITHA	23C01074	2026-03-30 19:57:20	Valid
53	YOGITHA	23C01074	2026-03-30 19:56:29	Valid
51	YOGITHA	23C01074	2026-03-30 19:55:44	Valid
51	YOGITHA	23C01074	2026-03-30 19:55:44	Valid
51	YOGITHA	23C01074	2026-03-30 19:55:44	Valid
51	YOGITHA	23C01074	2026-03-30 19:55:43	Valid
51	YOGITHA	23C01074	2026-03-30 19:55:43	Valid
51	YOGITHA	23C01074	2026-03-30 19:55:43	Valid
51	YOGITHA	23C01074	2026-03-30 19:51:00	Valid
51	YOGITHA	23C01074	2026-03-30 19:50:59	Valid

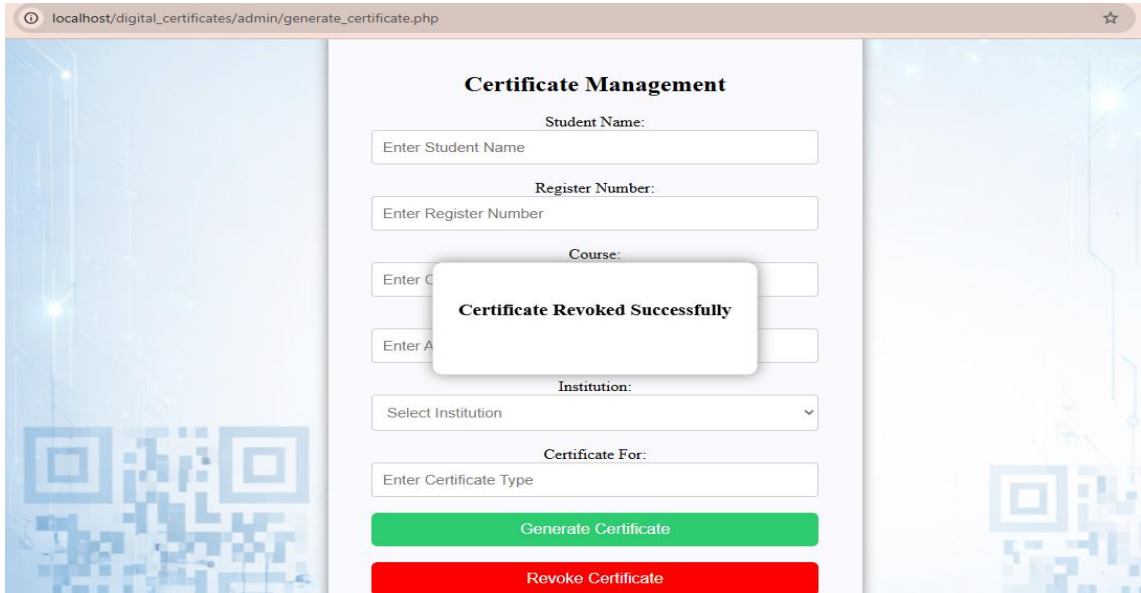
Fig 8.17

localhost/digital_certificates/admin/view_verification_logs.php

1	OLIVER SMITH	23BCA001	2026-03-08 19:50:54	Valid
37	JOSEPH HALL	24BBA001	2026-03-08 19:37:39	Tampered
34	EMILY WALKER	22BA0019	2026-03-08 19:34:50	Invalid
47	External Student	EXT1772977565	2026-03-08 19:28:10	Valid
46	External Student	EXT1772977565	2026-03-08 19:27:21	Valid
45	External Student	EXT1772977565	2026-03-08 19:26:53	Valid
42	MATTHEW LEE	22BA0018	2026-03-08 19:24:32	Valid
44	External Student	EXT1772977565	2026-03-08 19:23:41	Valid
37	JOSEPH HALL	24BBA001	2026-03-08 18:16:55	Tampered
39	ELIZABETH ALLEN	24BBA0010	2026-03-08 18:10:26	Tampered
35	EMILY WALKER	22BA0019	2026-03-08 17:51:48	Invalid
33	MATTHEW LEE	22BA0018	2026-03-08 17:51:48	Invalid
32	MATTHEW LEE	22BA0018	2026-03-08 17:51:48	Invalid
34	EMILY WALKER	22BA0019	2026-03-08 17:51:48	Invalid
36	JOSEPH HALL	23BBA001	2026-03-08 17:51:48	Invalid
41	SAMUEL YOUNG	23BCA0015	2026-03-08 17:51:48	Invalid
20	ALEXANDER ROBINSON	22BBA0015	2026-03-08 17:50:32	Valid
22	HARPER CLARK	22BC0015	2026-03-08 17:50:32	Valid

Fig 8.18

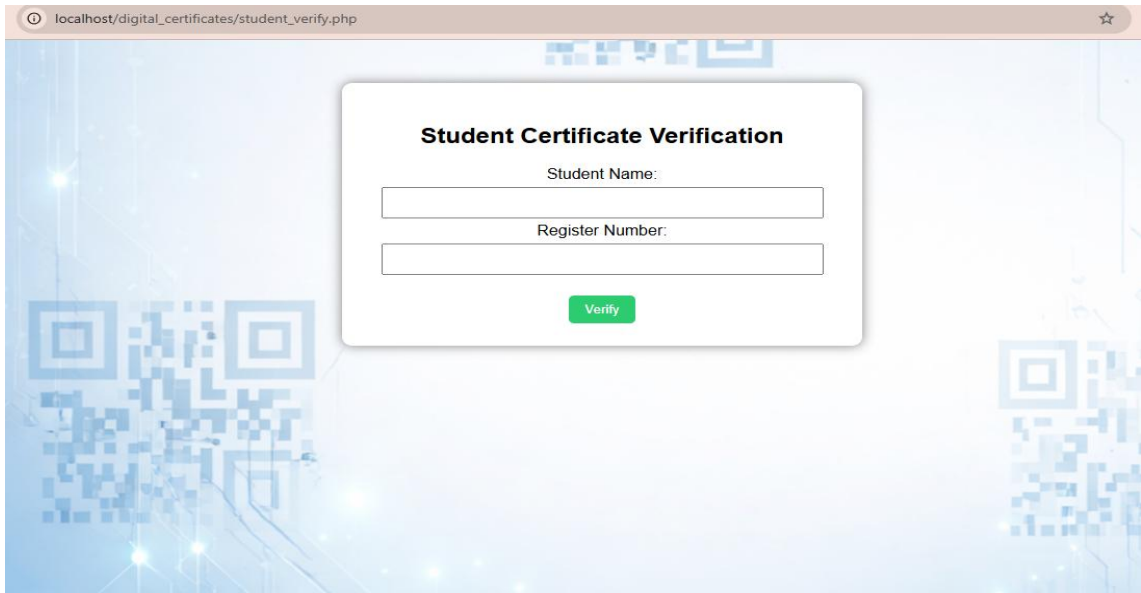
CERTIFICATE REVOCATION SUCCESS MESSAGE:



The screenshot shows a web browser window with the address bar displaying "localhost/digital_certificates/admin/generate_certificate.php". The page title is "Certificate Management". The form contains several input fields: "Student Name:" (with placeholder "Enter Student Name"), "Register Number:" (with placeholder "Enter Register Number"), "Course:" (with placeholder "Enter C"), and "Institution:" (with a dropdown menu labeled "Select Institution"). There are also two buttons at the bottom: "Generate Certificate" (green) and "Revoke Certificate" (red). A modal dialog box is displayed in the center of the form, containing the text "Certificate Revoked Successfully".

Fig 8.19

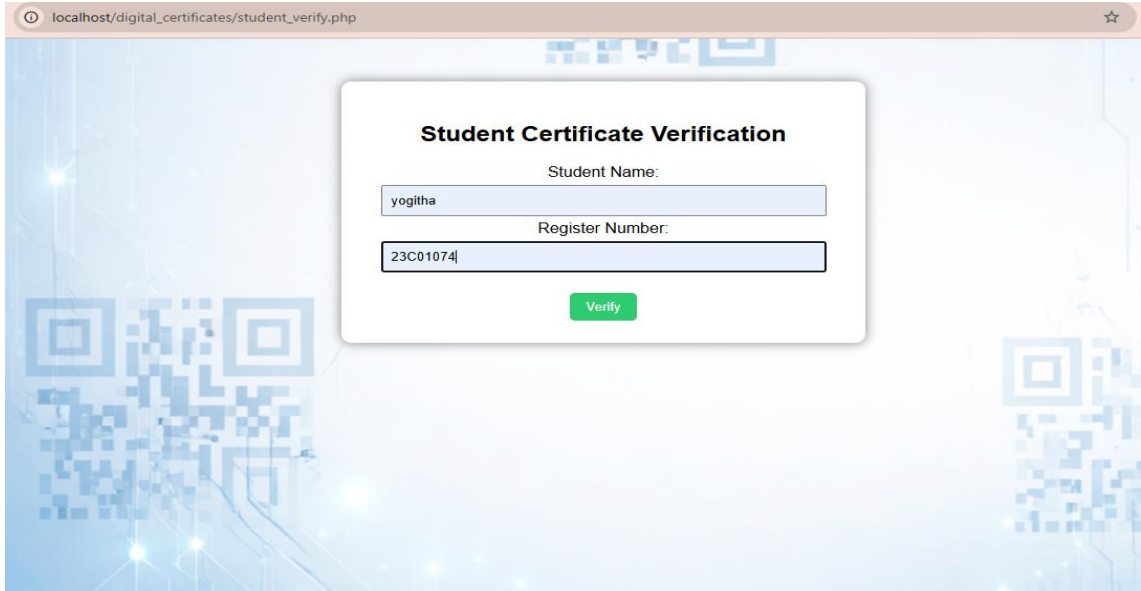
STUDENT CERTIFICATE VERIFICATION PAGE:



The screenshot shows a web browser window with the address bar displaying "localhost/digital_certificates/student_verify.php". The page title is "Student Certificate Verification". The form contains two input fields: "Student Name:" and "Register Number:". Below the input fields is a green button labeled "Verify".

Fig 8.20

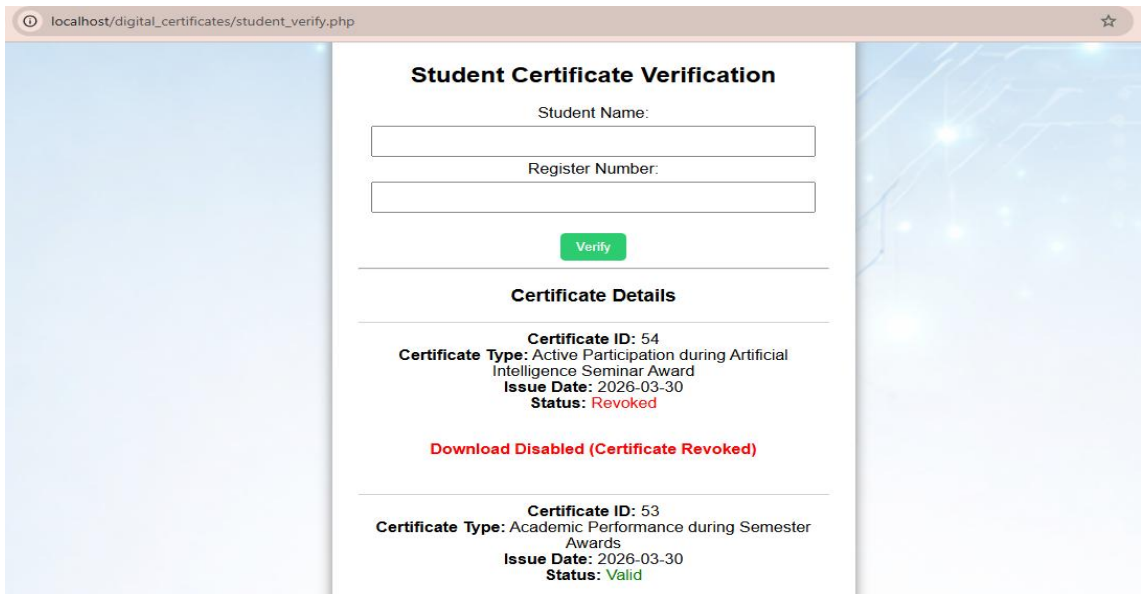
STUDENT DETAILS ENTERED FOR VERIFICATION:



The screenshot shows a web browser window with the URL `localhost/digital_certificates/student_verify.php`. The page has a light blue background with a QR code pattern. A central white box titled "Student Certificate Verification" contains two input fields. The first field, labeled "Student Name:", contains the text "yogitha". The second field, labeled "Register Number:", contains the text "23C01074". Below these fields is a green button labeled "Verify".

Fig 8.21

STUDENT VERIFICATION RESULT WITH CERTIFICATE SUMMARY:



The screenshot shows the same web browser window as Fig 8.21, but the form is now displaying the verification results. The "Student Name:" and "Register Number:" fields are empty. The "Verify" button is still present. Below the form, there is a section titled "Certificate Details" which lists two certificates. The first certificate has ID 54, Type "Active Participation during Artificial Intelligence Seminar Award", Issue Date "2026-03-30", and Status "Revoked". The second certificate has ID 53, Type "Academic Performance during Semester Awards", Issue Date "2026-03-30", and Status "Valid". A red message "Download Disabled (Certificate Revoked)" is displayed below the first certificate.

Certificate ID	Certificate Type	Issue Date	Status
54	Active Participation during Artificial Intelligence Seminar Award	2026-03-30	Revoked
53	Academic Performance during Semester Awards	2026-03-30	Valid

Fig 8.22

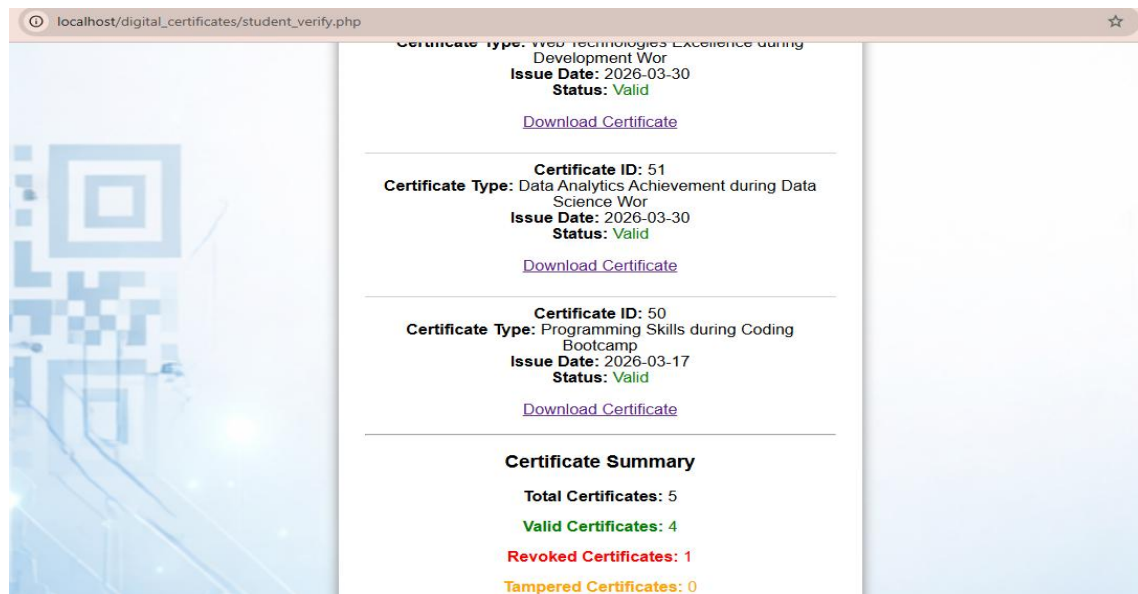


Fig 8.23

STUDENT VERIFICATION WITH TAMPERED CERTIFICATE DETECTION:

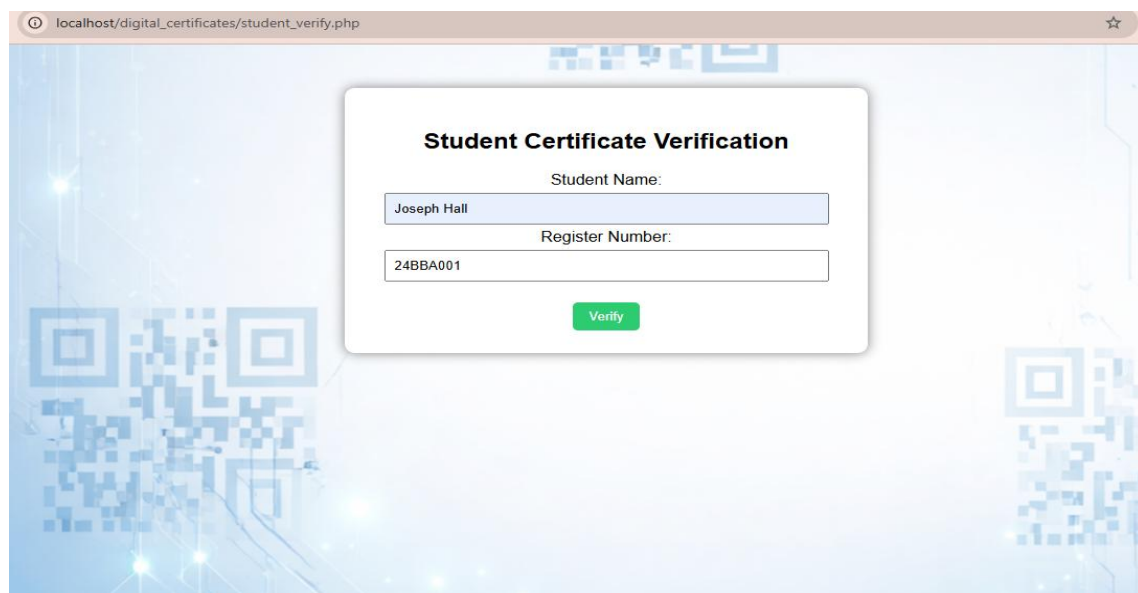


Fig 8.24

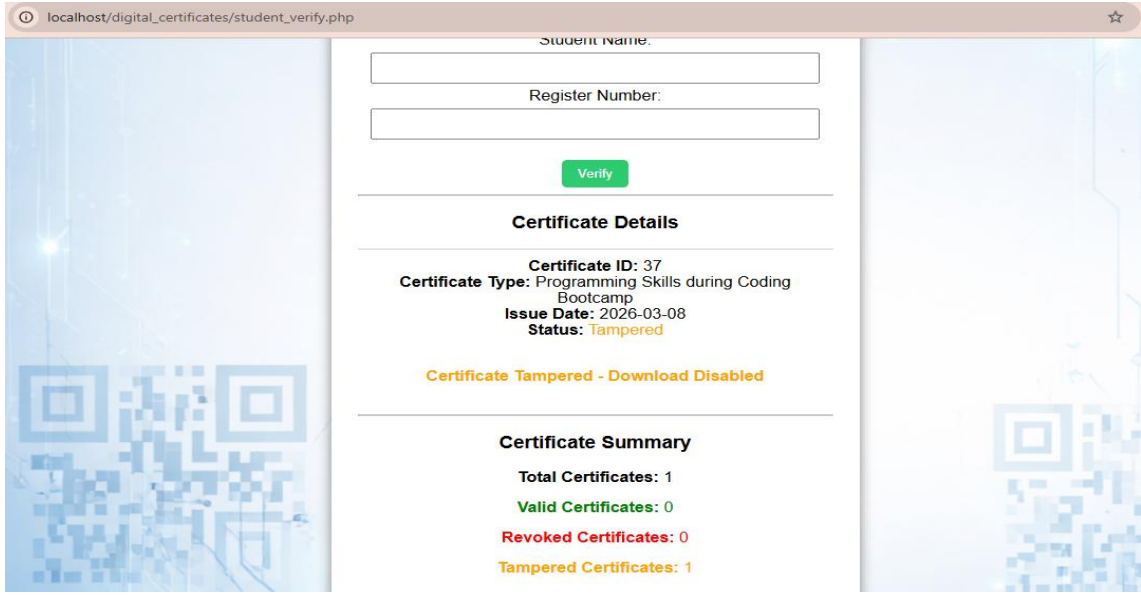


Fig 8.25

INVALID CERTIFICATE VERIFICATION RESULT:



Fig 8.26

CONCLUSION

9. CONCLUSION:

The project titled “**QR Code Based Digital Certificate Verification System to Prevent Forgery**” successfully provides a secure and efficient solution for managing and verifying digital certificates. The system addresses the major issues of certificate forgery, manual verification, and lack of transparency in traditional methods by introducing an automated and reliable verification mechanism.

The implementation of QR code technology enables quick and easy verification of certificates, while the use of hashing techniques ensures the integrity of certificate files by detecting any unauthorized modifications. The system also provides essential functionalities such as certificate generation, external certificate upload, certificate revocation, and verification logging, making it a complete solution for certificate management.

The developed system improves the overall efficiency of institutions by reducing manual effort and providing instant verification for students, organizations, and employers. Additionally, the inclusion of security features such as tamper detection and data masking enhances the reliability and trustworthiness of the system.

In conclusion, the proposed system effectively prevents certificate forgery, ensures data integrity, and simplifies the verification process. It can be further enhanced by deploying it on a live server and integrating advanced security features, making it suitable for real-world applications in educational institutions and organizations.

FUTURE ENHANCEMENTS

10. FUTURE ENHANCEMENTS:

- **Live Server Deployment**

The system can be deployed on a live server with a domain to enable access from anywhere instead of using a local IPv4 address.

- **Enhanced Security**

Advanced security measures such as encryption and multi-factor authentication can be implemented to improve system protection.

- **Multi-Institution Support**

The system can be extended to support multiple institutions for centralized certificate management.

- **Mobile Application Integration**

A mobile application can be developed for easier QR code scanning and certificate verification.

- **Notification System**

Email or SMS notifications can be added to inform users about certificate generation and updates.

- **Role-Based Access Control**

Different admin roles with specific permissions can be introduced for better system management.

- **Backup and Recovery**

Data backup and recovery mechanisms can be implemented to prevent data loss.

- **Advanced Data Security**

Additional security techniques can be used to protect stored certificate data.

- **Blockchain Integration**

Blockchain technology can be explored to provide highly secure and tamper-proof certificate storage.

BIBLIOGRAPHY

11. BIBLIOGRAPHY:

The development of the **QR Code Based Digital Certificate Verification System to Prevent Forgery** was supported by various resources, including books, online documentation, and technical references. The following sources were referred during the project development:

➤ BOOKS & RESEARCH PAPERS:

- Welling, Luke & Thomson, Laura. *PHP and MySQL Web Development*. Addison-Wesley, 2016.
- Nixon, Robin. *Learning PHP, MySQL & JavaScript*. O'Reilly Media, 2018.
- Elmasri, Ramez & Navathe, Shamkant. *Fundamentals of Database Systems*. Pearson, 2016.

➤ ONLINE REFERENCES & DOCUMENTATION:

- W3Schools: HTML, CSS, PHP, and MySQL tutorials – <https://www.w3schools.com>
- MySQL Documentation: Database management and SQL queries – <https://dev.mysql.com/doc/>
- PHP Official Documentation: Server-side scripting and functions – <https://www.php.net/docs.php>

➤ WEB-BASED LEARNING PLATFORMS:

- GeeksforGeeks: Programming tutorials and implementation guidance – <https://www.geeksforgeeks.org>
- Stack Overflow: Community support for debugging and development issues – <https://stackoverflow.com>

➤ LIBRARIES AND APIs USED:

- FPDF Library: PDF generation in PHP – <http://www.fpdf.org>
- FPDI Library: Importing and modifying PDF templates – <https://www.setasign.com/products/fpdfi>
- QR Code API: QR code generation service – <https://goqr.me/api>